

B o o k 1

Assignments in
**SENIOR
MATHS**

Grant Harrison



Photocopiable
blackline masters



B o o k 1

Assignments in
**SENIOR
MATHS**

Grant Harrison



Copying for educational purposes

The Australian Copyright Act 1968 (the Act) allows a maximum of one chapter or 10% of this book, whichever is the greater, to be copied by any educational institution for its educational purposes provided that the educational institution (or the body that administers it) has given a remuneration notice to the Copyright Agency Limited (CAL) under the Act.

For details of the CAL licence for educational institutions contact:

Copyright Agency Limited
Level 19, 157 Liverpool St
Sydney NSW 2000
Telephone: (02) 9394 7600
Facsimile: (02) 9394 7601
Email: info@copyright.com.au

Copying for other purposes

Except as permitted under the Act (for example, any fair dealing for the purposes of study, research, criticism or review) no part of this book may be reproduced, stored in a retrieval system, or transmitted in any form or by any means without prior written permission. All inquiries should be made to the publisher at the address below.

Copying of the blackline master pages

The purchasing educational institution and its staff are permitted to make copies of the pages marked as blackline master pages, beyond their rights under the Act, provided that:

- 1 the number of copies does not exceed the number reasonably required by the educational institution to satisfy its teaching purposes;
- 2 copies are made only by reprographic means (photocopying), not by electronic/digital means, and not stored or transmitted;
- 3 copies are not sold or lent;
- 4 every copy made clearly shows the footnote '© Grant Harrison and Learning Essentials 2003. This page may be photocopied for non-commercial classroom use.'

For those pages not marked as blackline master pages the normal copying limits in the Act, as described above, apply.

Project Manager: Shelley Barons
Design and formatting by *DiAn*
Printed in Australia by Ligare Book Printer



Ready to Use Resources

Instant Lessons
An imprint of Learning Essentials
A.B.N. 56 093 980 726
5 Como St
Malvern VIC 3144

Ph: (03) 9509 5366
www.learningessentials.com.au

ISBN 978 1 92209 088 1

This book has been written to provide comprehensive coverage of selected topics from the senior Mathematics syllabus. Book 1 concentrates on Algebra, beginning with the concepts of basic algebra, and progressing to more difficult topics, including factorisation, algebraic fractions, equations and inequations, and surds.

Each worksheet has been carefully graded, with the more difficult questions towards the end of each set. Every exercise has an example done to show the most common methods of working. The lessons should be able to be completed by students with minimal help from the teacher. Each unit has sufficient questions to occupy students for at least a 40-minute lesson.

Basic Algebra and Indices

1	Addition and subtraction	1
2	Multiplication and division	2
3	Substitution	3
4	Negative indices	4
5	Fractional indices	5
6	Distributive law	6
7	Binomial expansions	7
8	Basic algebra and indices review	8

Factorisation

9	Common factors	9
10	Difference of two squares	10
11	Grouping	11
12	Monic trinomials	12
13	Non-monic trinomials	13
14	Sum and difference of two cubes	14
15	Factorisation review	15

Algebraic Fractions

16	Simplifying algebraic expressions using factorisation	16
17	Multiplication and division of algebraic fractions	17
18	Addition and subtraction of algebraic fractions	18
19	Algebraic fractions review	19

Equations

20	One step and two step equations	20
21	Three step equations	21
22	Equations involving fractions	22
23	Exponential equations	23
24	Quadratic equations by factorisation	24
25	Quadratic equations using the quadratic formula or completing the square	25
26	Absolute value equations	26
27	Inequalities	27
28	Absolute value inequalities	28
29	Inequalities with the unknown in the denominator	29
30	Quadratic inequalities	30
31	Simultaneous equations – Elimination method	31
32	Simultaneous equations – Substitution method	32
33	Equations review	33

Surds

34	Simplifying surds	34
35	Addition and subtraction of surds	35
36	Multiplication of surds	36
37	Binomial products involving surds	37
38	Division of surds	38
39	Rationalising denominators	39
40	Surds review	40

Answers

41

Addition and subtraction

- Examples:**
- ① $4a + 2a = 6a$
 - ② $4a + 2b = 4a + 2b$
 - ③ $4a + 2 = 4a + 2$

Note: You can only add and subtract like terms.

Simplify:

1. $3a + 4b$

2. $6x + 5x$

3. $6c + 5$

4. $7x - 7$

5. $5a - a$

6. $2a^2 + 3a^2$

7. $a^2 - a$

8. $a^3 + a^2$

9. $4cg + 3gc$

10. $5x^2y - x^2y$

11. $4a^2c^3 - 2a^3c^2$

12. $\frac{1}{2}a^2 + 1\frac{1}{2}a^2$

13. $4a + 2b + 3a + 5b$

14. $6a - 3b + 4a$

15. $5a - 2b + a - 2b$

16. $-4c - 2a - c + a$

17. $a - b + a + b$

18. $3a^2 + 2a - a^2 - 7a$

19. $a^3 + 2a - a + 3a^3$

20. $4a - 2b - c + a$

21. $3a^2b + 4b^2a - 5ba^2 + ab^2$

22. $7ab - 7a - b + ba$

23. $12gh + 4g^2 - 2h + 4gh - h - 6g^2$

24. $\frac{2}{3}a^3 + \frac{1}{5}b^2 - \frac{5}{3}a^3 + \frac{4b^2}{5}$

25. $\frac{ab}{2} + \frac{ba}{2} - \frac{ca}{3} + \frac{4ca}{3}$

26. $8a^4 - 2a^2 + a^3 - 5a - a^4 + 3a^3 - 2a^2 + a$

27. $6a^2 - 6 + 4a - 7 - 5a^2 - 2$

28. $\pi r^2 + 4\pi r^2 - 2\pi r^2 + 2\pi r$

29. $5.2a^3 + 2.7 - 3a^3 - 0.2a^3 + 5.3$

30. $\frac{-y^2}{4} + \frac{3xy^2}{5} - \frac{x^2}{5} + \frac{3}{4}y^2 + \frac{2y^2x}{5} - \frac{9x^2}{5}$

Multiplication and division

- Examples:**
- ① $4a \times 5b = 20ab$
 - ② $2a^3 \times 6a^2 = 12a^5$
 - ③ $9c^5 \div 3c^2 = 3c^3$

Note: When multiplying expressions with the same base you **add** the indices.
When dividing expressions with the same base you **subtract** the indices.

Simplify:

- | | |
|------------------------------|---|
| 1. $a \times b$ | 16. $14c^3 \div 7c^3$ |
| 2. $a \times a \times a$ | 17. $2a^3 \times 4a^2 \times 5a$ |
| 3. $2c \times 4g$ | 18. $\frac{a^6}{a^4}$ |
| 4. $3a \times 2a$ | 19. $\frac{12a^8}{4a}$ |
| 5. $-5c \times 6$ | 20. $\frac{2c^5}{6c^2}$ |
| 6. $2a \times 3b \times 4c$ | 21. $5a^2b^3 \times 6a^4b^7$ |
| 7. $18a \div 9$ | 22. $7ab^2 \times 4a^3b$ |
| 8. $12b \div 3b$ | 23. $12a^6b^4 \div 6a^4b^7$ |
| 9. $4b \div 8$ | 24. $8c^2d^2 \div 2cd^2$ |
| 10. $6a \div 3b$ | 25. $\frac{1}{2}a^4b \times \frac{2}{3}ab$ |
| 11. $12a^2 \div 4a$ | 26. $0.3b^3 \div 0.01b^3$ |
| 12. $a^3 \times a^6$ | 27. $\frac{4a^3 \times 5a^2}{10a^4}$ |
| 13. $a^5 \div a^5$ | 28. $\frac{4c^2d^3 \times 3cd^6}{6cd^4}$ |
| 14. $2a^3 \times 3b^2$ | 29. $\frac{5ab \times 6a^2b^4}{45a^3b^3}$ |
| 15. $4a \times 5b \times 3a$ | 30. $\frac{2a^3b^7 \times 6a^4b^6}{3a^6b^4 \times 8ab^3}$ |

Substitution

Examples: If $a = 2$, $b = 3$ and $c = -1$, evaluate:

① $2a + 3b = 2 \times 2 + 3 \times 3 = 13$

② $3a^2 = 3 \times 2^2 = 3 \times 4 = 12$

③ $ab - 4c = 2 \times 3 - 4 \times -1 = 6 + 4 = 10$

Evaluate the following algebraic equations by substituting:

$a = 2$, $b = 4$, $c = -3$, $d = 1$, $e = 0$, $f = \frac{1}{2}$, $g = -1$, $h = -\frac{2}{3}$

1. ab

16. $\frac{b}{e}$

2. $b - c$

17. $gh + f$

3. $2f + 6g$

18. $\sqrt{b^2 + c^2}$

4. $2b^2$

19. $\frac{f}{h}$

5. $abde$

20. $4g^3 \times 3d^2$

6. h^2

21. $\frac{2}{3}f - \frac{c}{6}$

7. $\frac{3c}{g}$

22. $\frac{d}{a} \times \frac{g}{h}$

8. $5c - 6d$

23. $(3b + f)^2$

9. g^{57}

24. $(c - h)^2$

10. $b \div f$

25. $c(2a^2 + 3d)$

11. a^3g^4

26. $c^2 - 4c - 5$

12. $4f \div h$

27. $\frac{6a}{3c + 3d}$

13. h^4

28. $(f - g)(f + g)$

14. $\frac{ef}{d}$

29. $(4a^2 - c^2)(3b^2 + g^2)$

15. $3g - 4c$

30. $\frac{2b}{c} - \frac{g}{f} + \frac{d}{h}$

Negative indices

Examples: ① $a^{-4} = \frac{1}{a^4}$

② $3a^{-2} = \frac{3}{a^2}$

③ $\frac{a^4}{a^6} = a^{-2} = \frac{1}{a^2}$

④ $(a^{-2}b^3)^4 = a^{-8}b^{12} = \frac{b^{12}}{a^8}$

Note: $a^{-n} = \frac{1}{a^n}$

Simplify:

1. a^{-2}

16. $g^{-8} \div g^{-2}$

2. $2a^{-5}$

17. $(a^{-3}b^4)^2$

3. a^4b^{-2}

18. $(2c^{-2}b^3)^{-2}$

4. $7a^{-1}b$

19. $\left(\frac{1}{2}\right)^{-3}$

5. $a^3 \div a^5$

20. $\left(\frac{2}{a}\right)^{-2}$

6. $8a^2 \div 4a^3$

21. $10a^3b \div 5a^5b$

7. $(a^{-3})^2$

22. $4c^{-2}b^3 \div 8c^{-3}b^5$

8. $a^{-2} \times a^{-3} \times a^4$

23. $\frac{5a^2 \times 2a^{-6}}{10a^{-4}}$

9. $2a^{-4} \times 3a^{-1}$

24. $\left(\frac{5a^{-2}}{7a^{-4}}\right)^{-2}$

10. $\frac{b^6}{b^8}$

25. $\frac{4c^6 \times 5c^{-8}}{3c^{-2} \times 2c}$

11. $\frac{12c^9}{6c^{12}}$

26. $\frac{(a^{-2}b^3)^2}{(ab^4)^{-2}}$

12. $3a^{-2} \times 4a^2$

27. $(0.6)^{-2}$

13. $(2c^{-4})^3$

28. $\frac{4a^2b^{-3} \times 6a^{-3}b}{3a^{-5}b^2 \times 2a^{-1}b^3}$

14. $\frac{f^{-6}}{f^4}$

29. $(2a^3b^{-4})^2 \div (3a^{-1}b^{-2})^{-3}$

15. $\frac{c^{-3}}{c^{-7}}$

30. $\frac{(2a^{-2}b)^3 \times 4ab^{-3}}{8a^{-2}b^{-5} \times (4a^{-2}b)^{-1}}$

Fractional indices

Examples: ① $25^{\frac{1}{2}} = \sqrt{25} = 5$

② $8^{-\frac{1}{3}} = \frac{1}{8^{\frac{1}{3}}} = \frac{1}{\sqrt[3]{8}} = \frac{1}{2}$

③ $(a^{\frac{1}{2}})^{-3} \times (a^4)^{-\frac{1}{2}} = a^{-\frac{3}{2}} \times a^{-2} = a^{-\frac{7}{2}} = \frac{1}{a^{\frac{7}{2}}} = \frac{1}{\sqrt{a^7}}$

Note: $a^{\frac{m}{n}} = \sqrt[n]{a^m}$ or $(\sqrt[n]{a})^m$

Simplify, writing answers with positive indices or in radical form:

1. $36^{\frac{1}{2}}$

2. $8^{\frac{1}{3}}$

3. $21^{\frac{1}{4}}$ (correct to 2 decimal places)

4. $16^{-\frac{1}{4}}$

5. $27^{\frac{2}{3}}$

6. $64^{-\frac{5}{6}}$

7. $(\frac{1}{25})^{\frac{1}{2}}$

8. $(\frac{1}{49})^{-\frac{1}{2}}$

9. $(\frac{8}{27})^{\frac{1}{3}}$

10. $(\frac{16}{81})^{-\frac{1}{4}}$

11. $(\frac{32}{243})^{-\frac{1}{5}}$

12. $(\sqrt{a})^2$

13. $\sqrt{a} \times \sqrt[3]{a}$

14. $(\sqrt{b})^3 \div (\sqrt[3]{b})^2$

15. $a^{\frac{1}{3}} \times a^{\frac{1}{4}}$

16. $\frac{4c^{\frac{1}{2}}}{2c^2}$

17. $\sqrt[5]{b} \div b^{\frac{1}{5}}$

18. $(\sqrt{c})^3 \times \sqrt{c^3}$

19. $5\sqrt{a} \div 2a^{\frac{3}{2}}$

20. $(\sqrt{b})^{-3}$

21. $(c^{\frac{1}{4}})^{-\frac{1}{2}}$

22. $\frac{4b^{\frac{1}{2}} \times 2b^{\frac{1}{3}}}{16b^{\frac{1}{6}}}$

23. $\frac{\sqrt[3]{a^2} \times \sqrt{a^5}}{\sqrt[4]{a^6}}$

24. $(0.64)^{-\frac{1}{2}}$

25. $\sqrt[5]{37^2} \times 14.6$ (correct to 2 decimal places)

26. $\sqrt{a^3} \times \sqrt[3]{b} \times \sqrt[3]{a^6}$

27. $\frac{6b^{-2} \times 5c^{\frac{1}{2}}}{15\sqrt{bc}}$

28. $\frac{\sqrt{a^{-2}} \times \sqrt{16a^{-4}}}{\sqrt[3]{8a^6}}$

29. $\frac{10^{-1} \times 1000^{\frac{2}{5}}}{\sqrt{100} \times (1000)^{-\frac{1}{4}}}$

30. $\sqrt[3]{a^6 b^{-9} c^2} \div (ab^2 c)^{-1}$

Distributive law

Examples: Expand: ① $5(a + 2) = 5a + 10$

② $3(x - 2) - 2(x - 5) = 3x - 6 - 2x + 10 = x + 4$

Note: $a(b + c) = ab + ac$

Expand and collect like terms where necessary.

1. $6(a + 4)$

2. $3(2a - 5)$

3. $4(3a + 2b)$

4. $a(a + 3)$

5. $3b(7 - 2b^2)$

6. $4b^3(3b^2 - 2a^2)$

7. $-4(a^2 - 3)$

8. $-(1 - a)$

9. $6 - 4(a + 2)$

10. $5(2a + 3b - c)$

11. $5a^2(a - 3)$

12. $4bc(2b - 3c)$

13. $5(c - 2) - (c - 3)$

14. $a(a - b) + b(b - a)$

15. $7 - 2(a - 4)$

16. $-(2a - 3) - (4 - a)$

17. $5a^2 - a^2(a + 2)$

18. $4(2x + 3y) - (x + y) + (x - y)$

19. $-b(3 - b) + 3b$

20. $\frac{1}{2}(2a + 4b) + \frac{1}{3}(6a - 3b)$

21. $4(a + 2) + 2(a + 6)$

22. $5(2c - 7) + 2(c - 4)$

23. $4(a - 2) - 2(1 + a)$

24. $a(a^2 - 5a + 4) + 2(a^3 - a^2 - 2a)$

25. $2a(a - 3b - a^2) - (3a^2 + ab - 4a^3)$

26. $2(x + 2y) - 5(y - x) - 3(2x + y)$

27. $2a^2 - a(1 - 3a)$

28. $b(2a - 3b) + (3ab - b^2)$

29. $x + 6y - (3x + 2y) - (x - y)$

30. $a^2(2a - 1) + a(3a^2 - 4a) - 5(1 - a^2)$

Binomial expansions

Examples: ① $(a + 4)(a + 6) = a^2 + 4a + 6a + 24$
 $= a^2 + 10a + 24$

② $(a + 5)^2 = a^2 + 10a + 25$

③ $(a - 5)^2 = a^2 - 10a + 25$

④ $(a + 5)(a - 5) = a^2 - 25$

Note: $(a + b)(c + d) = ac + ad + bc + bd$

Expand and collect like terms where necessary:

1. $(a + 3)(a + 2)$

2. $(b - 5)(b + 7)$

3. $(c - 1)(c - 2)$

4. $(2a - 3)(a + 4)$

5. $(2a + 3b)(a + b)$

6. $(1 - 3a)(2a - 4)$

7. $(a - b)(b - a)$

8. $(a + 3)^2$

9. $(2a + 5)^2$

10. $(a - 7)^2$

11. $(b - 2a)^2$

12. $(5c + 2g)^2$

13. $(3b^2 - 2a)^2$

14. $(a - 2)(a + 2)$

15. $(3a + 5b)(3a - 5b)$

16. $(2c^3 - c^2)(c^2 + 2c^3)$

17. $(3a - 4b)^2$

18. $(a^2 - 2a)(a^2 + 2a)$

19. $(a^2 + 1)^2 - (a^2 - 1)^2$

20. $(a + 4)^2 + (a + 3)^2$

21. $(x - 2)(x + 6) + x^2 - 3x$

22. $(c + 4)(c + 6) - (c + 2)(c + 7)$

23. $(b - 4)^2 - (b - 2)(b + 3)$

24. $12 - 2(a - 3) - (2a + 6)$

25. $(2a - 5)^2 - (3a + 4)^2$

26. $(f - 6)(2f^2 + 3f + 1)$

27. $(2b^2 - 1)(b^2 - 3b + 4) - (1 - 3b^2)^2$

28. $(2a - b)^3$

29. $(a - 4)(a + 4) - (a - 2)(a - 1) + 5(a + 2)$

30. $(2b - 3a)^2 - (a - 2b)(a + 2b)$

Basic algebra and indices review

Simplify:

1. $5a - a$

2. $5a - 5$

3. $5a - 4b$

4. $4a - 3b + 2a - b$

5. $3a^2 - 4a - 2a^2 + 3a$

6. $3a^4 \times 2a^6$

7. $12b^6 \div 24b^2$

8. $(3a^2b)^3$

9. $\frac{6a^4b \times 3a^2b^3}{9a^5b^4}$

10. $12a^5 \div 4a^5$

11. $\frac{24a^4}{4a^8}$

12. $(1\frac{3}{4})^{-2}$

13. $\frac{5a^2b^2 \times 4ab^2}{2a^6b^2 \times 15ab}$

14. $\frac{4a^{-2} \times 3a^{-4}}{6a^{-3}}$

15. $(2c^2b^{-3})^2 \div (4a^{-2}b)^{-1}$

16. $8^{\frac{2}{3}}$

17. $128^{-\frac{5}{7}}$

18. $\frac{-1}{16^{\frac{3}{4}}}$

19. $(3\frac{3}{8})^{-\frac{2}{3}}$

20. $\sqrt{a^5} \times \sqrt[3]{a^6} \times \sqrt{a}$

21. $\frac{4a^{-2} \times 5a^3}{20a^2}$

22. $\sqrt[4]{16a^8} \div \sqrt[5]{32a^{10}}$

23. $\frac{\sqrt{a^{-6}} \times \sqrt{25a^{-4}}}{\sqrt[3]{125a^9}}$

If $a = 2$, $b = -1$ and $c = \frac{1}{2}$, evaluate:

24. $4abc^2$

25. $6c - ab$

26. $\frac{ab^5}{c}$

Expand and collect like terms:

27. $4(2a + 3) + 2(a - 5)$

28. $2a(3b - a) - a(b + a)$

29. $12a - 2a(4 - 3a)$

30. $(3a - 2b)^2$

31. $(5a + 3c)(5a - 3c)$

32. $(a - 4)^2 - (a + 4)^2$

33. $(a - 3)(a + 3) - (a + 1)(a + 4)$

34. $(2a + b)^2 - (a + 3b)(a - 2b)$

Common factors

Examples: Factorise:

① $8a + 12 = 4(2a + 3)$

② $a^5 + a^2 = a^2(a^3 + 1)$

③ $4(a + b) + x(a + b) = (a + b)(4 + x)$

Note: Take out the highest common factor.

Factorise:

1. $5a + 15$

2. $8a - 16$

3. $4a + 6b$

4. $3a + 6b + 12c$

5. $b^4 - b^6$

6. $8ac + 6ce$

7. $ax - bx + cx$

8. $2y^2 + 4y$

9. $x^2 - xy$

10. $4b^3 - 6b^2 - 8b$

11. $a(c + d) - at$

12. $35 - 56y$

13. $m - m^3$

14. $\frac{a}{2} + ab$

15. $x^2y + xy^2$

16. $a^3b^4 - a^2b^6$

17. $-15b^2 - 10b$

18. $3a^2 + a(b + c)$

19. $5c^2 - 15c$

20. $4c^2d^3 + 8cd^4$

21. $3(a + b) + c(a + b)$

22. $(x + a) + (x + a)^2$

23. $3x^2 + x(y + t)$

24. $9a^5 - 6a^3 + 12a^6$

25. $15x^3y^4 + 25xy^6$

26. $\frac{2a}{3} + \frac{5b}{6}$

27. $\sqrt{x} + x$

28. $(x + y)^2 + 4(x + y)^3$

29. $8x^2y^2z - 12x^2yz$

30. $10a^3b^4 - 25ab^6 + 30a^3b$

Difference of two squares

Examples: ① Factorise $x^2 - 25 = (x - 5)(x + 5)$

② Factorise $3a^2 - 27b^2 = 3(a^2 - 9b^2)$
 $= 3(a - 3b)(a + 3b)$

Note: Take out the highest common factor before using the difference of two squares.

Factorise:

1. $a^2 - 16$

2. $b^2 - 9$

3. $c^2 - 121$

4. $a^2 - b^2$

5. $1 - a^2$

6. $49 - x^2$

7. $9a^2 - 25$

8. $36b^2 - 49c^2$

9. $16 - 25a^2$

10. $5a^2 - 80$

11. $a^2b^2 - c^2$

12. $1 - 49m^2$

13. $ab^2 - a^3$

14. $8c^2 - 2b^2$

15. $x^3 - x$

16. $(1 + a)^2 - 9$

17. $162a^2 - 50c^2$

18. $a^4 - 49$

19. $\frac{x^2}{a^2} - \frac{y^2}{b^2}$

20. $3c^2d^2 - 48d^2$

21. $2a^2 - 0.18$

22. $81a^2 - 25c^2$

23. $48c^2 - 75b^2$

24. $(ab)^2 - c^2d^4$

25. $3(a + c)^2 - 27$

26. $(a + 5)^2 - (a - 2)^2$

27. $5a^2b^2 - 45b^4$

28. $m^4 - 1$

29. $4x^2 - 16(y + z)^2$

30. $5a^2b^4 - 80a^2c^4$

Grouping

Examples: ① Factorise: $ax + ab + cx + cb = (ax + ab) + (cx + cb)$
 $= a(x + b) + c(x + b)$
 $= (x + b)(a + c)$

② Factorise: $x^2 - x - ax + a = (x^2 - x) + (-ax + a)$
 $= x(x - 1) - a(x - 1)$
 $= (x - 1)(x - a)$

Factorise the following:

1. $ba + 2b + ca + 2c$

2. $3a(2b - 5c) - g(2b - 5c)$

3. $6a + 6b + ac + bc$

4. $5ab - 10b + a - 2$

5. $a^2 + a + ba + b$

6. $2xy + 3xc + 4gy + 6gc$

7. $8ab - 20ac + 4gb - 10cg$

8. $b^2 - b - cb + c$

9. $a^3 - 4a^2 + 2a - 8$

10. $x^3 - 5x^2 - 10x + 50$

11. $c^3 - 3c^2 - c + 3$

12. $6a - 3a^2 + 2b - ba$

13. $c^3 - 8 - 4c^2 + 2c$

14. $a^2b - ab^2 + 6a - 6b$

15. $x^4 - x^3 + ax - a$

16. $cd - ed + ce - e^2$

17. $15ab - 10at + 3gb - 2gt$

18. $ac + bd - ad - bc$

19. $ca + cd - 7a - 7d$

20. $10a - 120b + a^2 - 12ab$

21. $ab^2 + ac - b^2y - cy$

22. $x^3 - 2 - 2x + x^2$

23. $ab^2 - cb + abe - ce$

24. $3a^3 - 3a^2 + 3a - 3$

25. $a^2 - b^2 + 2a + 2b$

26. $4x^2 - 25y^2 - 2x - 5y$

27. $a - b - a^2 + b^2$

28. $\frac{ab}{2} - \frac{b}{4} + 10a - 5$

29. $40ab - 30a^2 - 60b^2 + 45ba$

30. $x^4 - a^4 + 2x - 2a$

Monic trinomials

Examples: ① Factorise: $x^2 + 6x + 8 = (x + 4)(x + 2)$

② Factorise: $2x^2 - 10x - 12 = 2(x^2 - 5x - 6)$
 $= 2(x - 6)(x + 1)$

Note: Take out the highest common factor before using the trinomial method.

Factorise:

1. $a^2 + 7a + 10$

2. $b^2 + 14b + 13$

3. $x^2 + 5x + 6$

4. $x^2 - 7x - 8$

5. $c^2 - 2c - 8$

6. $y^2 + 5y - 6$

7. $a^2 - 8a + 12$

8. $x^2 - 2x - 63$

9. $a^2 + 29a + 100$

10. $b^2 - 2b - 35$

11. $x^2 - 19x - 42$

12. $x^2 - 17x + 42$

13. $x^2 + x - 42$

14. $x^2 - 14x - 15$

15. $a^2 - 2a - 99$

16. $b^2 - 18b + 32$

17. $c^2 - c - 72$

18. $g^2 + 30g + 144$

19. $x^2 - 6x + 9$

20. $b^2 - 22b + 72$

21. $2b^2 - 4b - 6$

22. $3a^2 + 21a + 30$

23. $3x^2 + 21x - 24$

24. $ab^2 - 13ba + 36a$

25. $2x^2 + 148x + 288$

26. $c^3 + 11c^2 + 28c$

27. $x^2 - 2xy + y^2$

28. $a^2 - ab - 6b^2$

29. $7c^2 + 112bc - 252b^2$

30. $\frac{a^4}{2} - \frac{a^3}{2} - 10a^2$

Non-monic trinomials

Examples: ❶ Factorise: $2x^2 + 11x + 5 = (2x + 1)(x + 5)$

❷ Factorise: $12x^2 - 102x - 54 = 6(2x^2 - 17x - 9)$
 $= 6(2x + 1)(x - 9)$

Note: Take out the highest common factor before using the non-monic method. Our aim in senior mathematics is to **write down** the factors without the use of other methods (eg the cross (X) method).

Factorise:

1. $2a^2 + 15a + 7$

2. $3b^2 + 17b + 10$

3. $2a^2 - 11a + 15$

4. $5c^2 - 39c - 8$

5. $3a^2 + 8a - 16$

6. $5a^2 + 3a - 2$

7. $3x^2 + 13x + 4$

8. $10a^2 + 15a - 10$

9. $2b^2 - 9b + 4$

10. $6a^2 + 26a - 20$

11. $11c^2 + 25c + 6$

12. $7a^2 + 2a - 5$

13. $9x^2 - 51x + 30$

14. $2 - x - x^2$

15. $10x^2 - 13x - 3$

16. $6a^2 - 19a + 10$

17. $5T^2 + 6T + 1$

18. $-3a^2 + 11a + 4$

19. $9a^2 + 30a + 25$

20. $18c^2 - 9c - 2$

21. $24b^2 - 28b + 8$

22. $5a^2 - 34ab - 7b^2$

23. $9a^2 + 24a + 16$

24. $12x^2 - 31x + 20$

25. $5x^2 + 9x - 72$

26. $30p^2 - 84p - 18$

27. $2 + a - 10a^2$

28. $12x^2 - 52xy - 40y^2$

29. $8c^3 - 18c^2 - 5c$

30. $\frac{7a^2}{2} - 10a + 6$

Sum and difference of two cubes

Examples: Factorise:

$$\textcircled{1} a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$\textcircled{2} a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$\begin{aligned}\textcircled{3} x^3 - 8 &= x^3 - 2^3 \\ &= (x - 2)(x^2 + 2x + 4)\end{aligned}$$

$$\begin{aligned}\textcircled{4} 3b^3 + 192 &= 3(b^3 + 64) \\ &= 3(b^3 + 4^3) \\ &= 3(b + 4)(b^2 - 4b + 16)\end{aligned}$$

Note: Take out the highest common factor before using the 'sum or difference of two cubes' method.

The middle term in the second, or longer, bracket is the **product of the numbers NOT** twice the product.

Factorise:

1. $x^3 + 1$

2. $x^3 - 1$

3. $a^3 + 27$

4. $a^3 - 125$

5. $2x^3 - 54$

6. $216 - b^3$

7. $8y^3 + 27$

8. $3a^3 + 375$

9. $a^3 - 64b^3$

10. $7m^3 - 7$

11. $ba^3 + c^3b$

12. $\pi R^3 - \pi r^3$

13. $64c^3 + 8b^3$

14. $x^5 + 8x^2$

15. $(a + b)^3 - a^3$

16. $x^6 - y^6$

17. $54a^3 + 16$

18. $1000c^3 - 343$

19. $(2a - 3b)^3 - (3b - 2a)^3$

20. $\frac{x^3}{8} - 8$

21. $\frac{a^3}{27} + \frac{b^3}{64}$

22. $b^3 - c^3 + b - c$

23. $27y^3 + 64a^3$

24. $b^3 + 8a^3 - cb - 2ca$

25. $0.001 - 0.008b^3$

26. $64 - (4 - x)^3$

27. $\frac{8}{a^3} + \frac{216}{b^3}$

28. $\frac{4}{3}\pi R^3 - \frac{500\pi}{3}$

29. $a^3 - m^6n^9$

30. $8x^5 - 32x^3 + x^2 - 4$

Factorisation review

Factorise:

1. $8a + 12$

2. $b^3 - b^4$

3. $c^2 - 25$

4. $x^2 + 8x + 12$

5. $ab + ac - b - c$

6. $3a^2 - 20a - 7$

7. $a^3 - 1000$

8. $15b^2 - 60$

9. $6x^2 + 32x + 10$

10. $x(a - b) + y(a - b)$

11. $x^3 - x^2 + ax - a$

12. $25x^2 - 45x + 20$

13. $\frac{y^2}{16} - \frac{x^2}{4}$

14. $8a^3b^2 - 10a^2b^4 + 4ba^5$

15. $8a^3 + 512$

16. $a^2 - a + 1 - a$

17. $8x^3 - 27y^6$

18. $x^2 - 2xy + y^2$

19. $12a^2 + 14a - 6$

20. $3c^2 - 63c - 216$

21. $2b^2 - 0.08$

22. $x^2 - 25 - ax - 5a$

23. $15 + 26x - 21x^2$

24. $125 - 20b^2$

25. $1 - a^2 + ab - a^3b$

26. $y(a - b) + 5(b - a)$

27. $21c^2 + 14bc - 35b^2$

28. $a^4b^2 - c^6$

29. $72a^2 + 68a - 60$

30. $2\pi R^2 + 2\pi RH$

31. $c^3 - c^2d - cd^2 + d^3$

32. $6a^3 + 6$

33. $(a + 3)^2 - 4(y - 2)^2$

34. $b^2a^9 - b^2$

35. $(a^2 - 4a + 4) - (a + 1)^2$

36. $8x^4 - 12x^3 + 40x^2 - 60x$

Simplifying algebraic expressions using factorisation

Note: You can only cancel algebraic terms when there is a **multiplication sign** between the terms (not addition or subtraction). When factorising algebraic expressions, you produce a multiplication sign.

Examples: Simplify:

$$\textcircled{1} \frac{3x+9}{3} \neq 3x+3$$

$$\neq x+9$$

$$\frac{3x+9}{3} = \frac{\cancel{3} \times (x+3)}{\cancel{3}} \quad \text{Factorisation}$$

$$= x+3$$

$$\textcircled{2} \frac{4a+8}{a+2} = \frac{4 \times (a+2)}{a+2}$$

$$= 4$$

$$\textcircled{3} \frac{a^2-25}{2a-10} = \frac{(\cancel{a-5})(a+5)}{2(\cancel{a-5})}$$

$$= \frac{a+5}{2}$$

$$\textcircled{4} \frac{5a-10}{2-a} = \frac{5(a-2)}{-1(a-2)}$$

$$= -5$$

Simplify:

$$1. \frac{7a+14}{7}$$

$$2. \frac{8c-10}{2}$$

$$3. \frac{5a+5b}{a+b}$$

$$4. \frac{6c-4T}{12c-8T}$$

$$5. \frac{x^2-9}{4x+12}$$

$$6. \frac{a^2+ab}{ab+b^2}$$

$$7. \frac{x^2-5x-6}{x^2-1}$$

$$8. \frac{3x^2-xy}{3xy-y^2}$$

$$9. \frac{1-9a^2}{a+3a^2}$$

$$10. \frac{3x-12}{8-2x}$$

$$11. \frac{(a+b)(b-c)}{2(a+b)}$$

$$12. \frac{(x+4)^2}{x+4}$$

$$13. \frac{3a^2+6a}{a^2+4a+4}$$

$$14. \frac{1-x^2}{x^2-1}$$

$$15. \frac{fg}{f^2g^2-fg}$$

$$16. \frac{(a-b)^2}{a^2-b^2}$$

$$17. \frac{x^2-7x+10}{x^2-25}$$

$$18. \frac{4a+8b+10c}{12a+24b+30c}$$

$$19. \frac{2x^3-16}{6x-12}$$

$$20. \frac{5x^2y^3-10x^3y^2}{2x^3y^2-4x^4y}$$

$$21. \frac{a^2-a}{a^2+a}$$

$$22. \frac{x^2-x-12}{x^2+4x+3}$$

$$23. \frac{4x^2-12x+9}{8x^3-27}$$

$$24. \frac{1-a^4}{2a-2}$$

$$25. \frac{a^2-ab+b-a}{a^2-b^2}$$

$$26. \frac{-2+3x-x^2}{1-x}$$

$$27. \frac{a^2+ab-2b^2}{ab-b^2}$$

$$28. \frac{x^3-x^4}{x-x^4}$$

$$29. \frac{(a+c)^2-m^2}{4a+4c-4m}$$

$$30. \frac{6(2a^2-8)}{4a^2-4a-8}$$

Multiplication and division of algebraic fractions

Examples: Simplify:

$$\begin{aligned} \textcircled{1} \quad & \frac{4x-8}{a^2-4} \times \frac{2a+4}{x^2-x-2} \\ &= \frac{4(\cancel{x-2})}{(a-2)(\cancel{a+2})} \times \frac{2(\cancel{a+2})}{(\cancel{x-2})(x+1)} \\ &= \frac{8}{(a-2)(x+1)} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad & \frac{a^2-b^2}{a^2-5a-6} \div \frac{4a+4b}{2a-12} \\ &= \frac{a^2-b^2}{a^2-5a-6} \times \frac{2a-12}{4a+4b} \\ &= \frac{(a-b)(\cancel{a+b})}{(\cancel{a+6})(a+1)} \times \frac{\cancel{2}(a-6)}{\cancel{4}(\cancel{a+b})} \\ &= \frac{a-b}{2(a+1)} \end{aligned}$$

Simplify:

$$1. \quad \frac{8a+6}{15} \times \frac{5}{4a+3}$$

$$2. \quad \frac{5a-5b}{12} \times \frac{6}{10a-10b}$$

$$3. \quad \frac{a-2}{a+2} \times \frac{3a+6}{a^2+2a-8}$$

$$4. \quad \frac{5}{b-2} \div \frac{10}{b^2-4}$$

$$5. \quad \frac{x^2-9}{5x+10} \div \frac{2x+6}{ax+2a}$$

$$6. \quad \frac{3a^2-ab}{ab} \times \frac{a^2b}{3ab-b^2}$$

$$7. \quad \frac{5}{x^3+64} \div \frac{10}{x+4}$$

$$8. \quad \frac{a-b}{a^3b^2} \times \frac{a^4b}{2b-2a}$$

$$9. \quad \frac{x^2-4x-5}{a^2-9} \div \frac{(x-5)^2}{4a-12}$$

$$10. \quad \frac{n^3+n^2}{a^2-a} \times \frac{a-a^3}{2n+2}$$

$$11. \quad \frac{c^2-2c-3}{c^2-11c-12} \times \frac{c^2-144}{c^2-c-6}$$

$$12. \quad \frac{(c-e)^2}{5} \times \frac{10}{c^2-e^2}$$

$$13. \quad \frac{x^2-y^2}{a^2-b^2} \times \frac{(a-b)^2}{(x+y)^2}$$

$$14. \quad \frac{3+2b-b^2}{2b+2} \times \frac{6}{3b-9}$$

$$15. \quad \frac{2a+6b}{4c-12} \times \frac{c^2-9}{12b+4a} \times \frac{8}{3+c}$$

$$16. \quad \frac{a^3-b^3}{c^2-9d^2} \div \frac{a^2-b^2}{2c-6d}$$

$$17. \quad \frac{x^2-2xy+y^2}{3c} \div \frac{2y-2x}{6c^2}$$

$$18. \quad \frac{(a+h)^2-a^2}{h}$$

$$19. \quad \frac{b^3-27}{2a^3+16} \times \frac{a^2-2a+4}{b^2-9}$$

$$20. \quad \frac{a^2-a-20}{b^2-4b} \times \frac{4}{(a-5)(b+4)} \div \frac{8+2a}{b^2-16}$$

Addition and subtraction of algebraic fractions

Note: Use the same rules as you would in arithmetic fractions.

Examples:

$$\textcircled{1} \frac{2a}{5} + \frac{3a}{10} = \frac{4a}{10} + \frac{3a}{10} = \frac{7a}{10}$$

$$\textcircled{2} \frac{5}{4b} - \frac{2}{6b} = \frac{15}{12b} - \frac{4}{12b} = \frac{11}{12b}$$

$$\textcircled{3} \frac{5}{x-2} + \frac{3}{x+5} = \frac{5(x+5)}{(x-2)(x+5)} + \frac{3(x-2)}{(x-2)(x+5)}$$

$$= \frac{5x+25+3x-6}{(x-2)(x+5)}$$

$$= \frac{8x+19}{(x-2)(x+5)}$$

$$\textcircled{4} \frac{5}{x^2-16} - \frac{2}{x^2-3x-4} = \frac{5}{(x-4)(x+4)} - \frac{2}{(x-4)(x+1)}$$

$$= \frac{5(x+1)}{(x-4)(x+4)(x+1)} - \frac{2(x+4)}{(x-4)(x+4)(x+1)}$$

$$= \frac{5x+5-2x-8}{(x-4)(x+4)(x+1)}$$

$$= \frac{3x-3}{(x-4)(x+4)(x+1)}$$

Simplify:

1. $\frac{3a}{2} + \frac{a}{3}$

2. $\frac{4a}{5} - \frac{a}{10}$

3. $\frac{a}{2} - \frac{a}{5}$

4. $\frac{a}{b} + \frac{c}{b}$

5. $\frac{4}{3a} - \frac{5}{2a}$

6. $\frac{4}{a} + \frac{5}{ab}$

7. $\frac{x+4}{3} + \frac{2x-1}{5}$

8. $\frac{a-2}{4} - \frac{a-3}{6}$

9. $\frac{5}{a^2b} + \frac{3}{ab^2}$

10. $\frac{a}{3} + \frac{4a}{5} - \frac{a}{4}$

11. $\frac{a-3b}{2} - \frac{3a+b}{6}$

12. $\frac{a+2}{6a} + \frac{a-5}{4a}$

13. $\frac{1}{ab} + \frac{4}{bc} - \frac{2}{ac}$

14. $\frac{4}{x-3} + \frac{3}{2x+1}$

15. $\frac{1}{x-y} - \frac{1}{x+y}$

16. $\frac{5}{2x-6} + \frac{1}{x-3}$

17. $\frac{5}{a-2} + \frac{2}{a^2-4}$

18. $\frac{5}{x^2y^3} - \frac{2}{x^3y^2}$

19. $\frac{4}{c^2-9} - \frac{2}{c^2-2c-3}$

20. $\frac{1}{x^2+xy} + \frac{2}{x}$

21. $\frac{3}{(a-b)^2} - \frac{4}{a-b}$

22. $\frac{5}{a^2-b^2} - \frac{2}{a-b}$

23. $\frac{4}{2x^2-2} + \frac{3}{5x-5}$

24. $\frac{4}{9-x^2} - \frac{2}{x-3}$

25. $\frac{5}{a-5} + \frac{2}{a+5} - \frac{3}{a^2-25}$

26. $a + \frac{5}{a-2}$

27. $\frac{b+1}{1-b} + \frac{b-1}{1+b}$

28. $\frac{x}{x^2-16} - \frac{2x}{4+x}$

29. $\frac{5}{(a-b)^2} - \frac{3}{b-a}$

30. $\frac{1}{x^3-1} + \frac{2}{x^2-1}$

Algebraic fractions review

Simplify:

1. $\frac{12a + 24}{6}$

2. $\frac{10a - 5b}{4a - 2b}$

3. $\frac{(a + b)^2}{a + b}$

4. $\frac{9 - x^2}{x - 3}$

5. $\frac{x^2 - 4x - 5}{2x - 10}$

6. $\frac{(x + 3)^2}{x^2 - 9}$

7. $\frac{2x^2 - 12x + 16}{4x^2 - 16}$

8. $\frac{15a^6b^3 - 10a^4b^7}{9a^8b - 6a^6b^5}$

9. $\frac{10a - 20b}{b + c} \times \frac{2b + 2c}{5a - 10b}$

10. $\frac{a^2 + 2ab + b^2}{(a + b)^2}$

11. $\frac{9 - x^2}{a^2 + a} \div \frac{x + 3}{1 + a}$

12. $\frac{(a + b)^2 - a^2}{2ab} \times \frac{a^2}{12a + 6b}$

13. $\frac{1 - x^3}{(a + b)^2} \div \frac{x^3 - 1}{2a + 2b}$

14. $\frac{a + b}{c - d} \times \frac{d - c}{b + a} \times \frac{c + d}{a - b}$

15. $\frac{(x^2 - 2x + 1)^2}{[x(x - 1)]^3}$

16. $\frac{2a}{5} + \frac{4a}{15}$

17. $\frac{a}{b} - \frac{b}{a}$

18. $\frac{2a - 3}{5} - \frac{a + 2}{4}$

19. $\frac{7}{6a} + \frac{2}{4a}$

20. $\frac{5}{c^2d^2} + \frac{3}{cd^3}$

21. $\frac{5}{x - 2} - \frac{2}{1 + x}$

22. $\frac{4}{x + 6} - \frac{3}{2x + 12}$

23. $\frac{2}{(x - 1)^2} + \frac{3}{x - 1}$

24. $\frac{5}{x + 2} - \frac{3}{(x + 2)(x - 4)}$

25. $\frac{3}{x^2 - 16} - \frac{4}{x + 4}$

26. $\frac{3b + 1}{3b - 1} - \frac{3b - 1}{3b + 1}$

27. $\frac{4x}{x^2 - 25} + \frac{3}{x + 5}$

28. $\frac{7}{4x} - \frac{2}{x^2 - 4x}$

29. $\frac{a}{a^2 - ab} - \frac{1}{a}$

30. $\frac{x + 3}{x^3 - 27} - \frac{5}{x^2 - 9}$

One step and two step equations

Examples: Solve for x :

① $x + 15 = 32$

$$x = 17$$

② $3x - 2 = 11$

$$3x = 13$$

$$x = 4\frac{1}{3}$$

③ $\frac{5x}{6} = -2$

$$5x = -12$$

$$x = -2\frac{2}{5}$$

Solve for a :

1. $a + 12 = 20$

2. $a - 11 = 11$

3. $7a = 56$

4. $\frac{a}{8} = 6$

5. $a + 223 = 315$

6. $a - 74 = 102$

7. $7a = 24$

8. $\frac{a}{10} = 10$

9. $2a + 5 = 13$

10. $3a - 4 = 7$

11. $\frac{5a}{2} = 4$

12. $\frac{9a}{4} = 2$

13. $6a + 8 = 12$

14. $5a + 12 = 3$

15. $4 + 3a = -9$

16. $-2a + 3 = 9$

17. $5 - 3a = 2$

18. $\frac{-4a}{5} = -6$

19. $5 - a = -2$

20. $7 = 2 + 3a$

21. $1 = \frac{3a}{2}$

22. $-4 - 5a = -2$

23. $\frac{a}{6} = \frac{1}{2}$

24. $7a = 5a + 3$

25. $\frac{8a}{12} = \frac{1}{3}$

26. $1 - a = -1$

27. $4a - 2.2 = 6.8$

28. $\frac{5a}{8} = 0.1$

29. $4a = 7a - 10$

30. $\frac{a}{3} + 5 = 7$

Three step equations

Examples: Solve for x :

$$\textcircled{1} \quad \frac{3x+2}{5} = 6$$

$$3x+2=30$$

$$3x=28$$

$$x=9\frac{1}{3}$$

$$\textcircled{3} \quad 8x+4=3x-7$$

$$8x=3x-11$$

$$5x=-11$$

$$x=-2\frac{1}{5}$$

$$\textcircled{2} \quad 5(2x-3)=4$$

$$10x-15=4$$

$$10x=19$$

$$x=1\frac{9}{10}$$

Note: The third example can be reduced to two steps.

Solve for a :

(Note: some of the later questions may require four steps)

$$1. \quad \frac{4a+3}{5} = 2$$

$$2. \quad \frac{5a-4}{2} = 9$$

$$3. \quad \frac{8a-1}{2} = 3$$

$$4. \quad \frac{6-2a}{4} = 5$$

$$5. \quad \frac{3-a}{2} = 5$$

$$6. \quad \frac{3a+1}{2} = -4$$

$$7. \quad \frac{7-2a}{4} = -3$$

$$8. \quad 4(3a-2) = 5$$

$$9. \quad 6(4a+3) = 22$$

$$10. \quad -2(a-4) = 6$$

$$11. \quad 9 = 3(2a+1)$$

$$12. \quad -4 = -(1+a)$$

$$13. \quad 7a-3 = 4a+6$$

$$14. \quad 8a+7 = 2a+9$$

$$15. \quad 9a+6 = 2-3a$$

$$16. \quad a-2 = 1-a$$

$$17. \quad 4-5a = 2-3a$$

$$18. \quad -2-3a = -6-a$$

$$19. \quad 6+2a = 7a$$

$$20. \quad 5(2a+3) = 3a-4$$

$$21. \quad 4-3a = -2(3a-1)$$

$$22. \quad 5(2a-4) = 2(3+a)$$

$$23. \quad \frac{5a-3}{4} = \frac{1}{2}$$

$$24. \quad 4(a+2) + 2(a+5) = 6$$

$$25. \quad 5(3a-1) - 2(a+1) = 6$$

$$26. \quad 5-3(a-2) = -4$$

$$27. \quad 6(5-a) - (1-a) = 2$$

$$28. \quad 3(a+2) + 2(3a+1) = 5(a+4)$$

$$29. \quad 5(a-1) - 3(a-2) = 3-5a$$

$$30. \quad 4(1-2a) - (3a+4) = 2(a+5) + (1-a)$$

Equations involving fractions

Examples: ① Solve for x : $\frac{3x-1}{2} - \frac{2x+5}{5} = 3$

Method 1

$$\frac{3x-1}{2} - \frac{2x+5}{5} = 3$$

$$\frac{15x-5}{10} - \frac{4x+10}{10} = 3$$

$$\frac{15x-5-4x-10}{10} = 3$$

$$\frac{11x-15}{10} = 3$$

$$11x-15 = 30$$

$$11x = 45$$

$$x = 4\frac{1}{11}$$

Method 2

$$\frac{3x-1}{2} - \frac{2x+5}{5} = 3$$

Multiply both sides of the equation by 10.

$$10 \times \frac{3x-1}{2} - 10 \times \frac{2x+5}{5} = 3 \times 10$$

$$5(3x-1) - 2(2x+5) = 30$$

$$15x-5-4x-10 = 30$$

$$11x-15 = 30$$

$$11x = 45$$

$$x = 4\frac{1}{11}$$

② Solve for x : $\frac{5}{3x} = \frac{4}{5} - \frac{2}{x}$

Multiply both sides by $15x$ (LCM of $3x$, 5 , x)

$$\frac{5}{3x} \times 15x = \frac{4}{5} \times 15x - \frac{2}{x} \times 15x$$

$$25 = 12x - 30$$

$$55 = 12x$$

$$x = 4\frac{7}{12}$$

Solve for a :

1. $\frac{3a-5}{4} = 7$

2. $\frac{2a}{3} + \frac{4}{5} = 2$

3. $\frac{3a}{4} - 5 = 6$

4. $5 - \frac{3a}{2} = 4$

5. $7 + \frac{2a}{3} = \frac{1}{6}$

6. $\frac{4a}{3} + 2 = a + \frac{3}{4}$

7. $\frac{2a+3}{4} + \frac{a+2}{5} = 2$

8. $\frac{3a-1}{6} + \frac{a-3}{9} = -1$

9. $\frac{4a-1}{5} - \frac{a-2}{3} = 4$

10. $5 - \frac{3a-2}{4} = 3$

11. $6 + \frac{2a-3}{4} = \frac{3}{8}$

12. $\frac{3a+5}{2} = \frac{a}{3}$

13. $\frac{5a-3}{2} = \frac{2a+1}{4}$

14. $\frac{a+2}{3} = 1 - \frac{a}{3}$

15. $\frac{3(a-2)}{5} + \frac{2(a+4)}{3} = 1$

16. $\frac{4(a-1)}{7} = \frac{3(a+2)}{6} - \frac{1}{14}$

17. $3a - \frac{a+4}{5} = -2$

18. $\frac{a+4}{6} = \frac{2a-3}{9} - \frac{a+1}{3}$

19. $\frac{4}{a+3} = 2$

20. $\frac{3}{a} + \frac{5}{2a} = 1$

21. $\frac{7a-3}{4} = \frac{a-2}{6}$

22. $\frac{7}{3a} = \frac{9}{2} - \frac{5}{a}$

23. $\frac{a-2}{a+4} = \frac{-2}{3}$

24. $\frac{a-3}{a+4} = \frac{a+1}{a-4}$

25. $\frac{a+2}{a-5} = 7$

26. $\frac{6}{a} = \frac{2}{a-7}$

27. $\frac{5}{a} - \frac{3}{a+2} = \frac{4}{a}$

28. $\frac{1}{a-1} + \frac{3}{2a-1} = 0$

29. $\frac{2}{a+1} - \frac{3}{a-2} = \frac{1}{a^2-a-2}$

30. $\frac{2}{a-3} = \frac{3}{a+3} - \frac{5}{a^2-9}$

Exponential equations

Examples: Solve for x :

① $2^x = 8$

$$2^x = 2^3$$

$$x = 3$$

② $9^{3x-1} = 27$

$$(3^2)^{3x-1} = 3^3$$

$$3^{6x-2} = 3^3$$

$$6x - 2 = 3$$

$$6x = 5$$

$$x = \frac{5}{6}$$

Solve for a :

1. $2^a = 16$

2. $3^a = 27$

3. $5^a = 625$

4. $7^{2a} = 49$

5. $6^{3a} = 216$

6. $2^{a+1} = 32$

7. $3^a = 1$

8. $\frac{2^{3a-1}}{4} = 1$

9. $16^{a+2} = 8$

10. $4^{2-a} = 32$

11. $3^a = \frac{1}{9}$

12. $5^{a+2} = \frac{1}{125}$

13. $2^{-a} = \frac{1}{128}$

14. $(\sqrt{2})^a = 16$

15. $3^a + 8 = 89$

16. $7^a = \frac{1}{49}$

17. $25^{1+a} = 125$

18. $(\sqrt{3})^a = 243$

19. $\frac{3^{a-1}}{9} = \sqrt{27}$

20. $\left(\frac{2}{3}\right)^a = 1.5$

21. $4^a + 3 \times 4^a = 8$

22. $4^{a-1} = \frac{1}{8}$

23. $3^{3a+2} = \sqrt{27}$

24. $5^{1-a} = 25\sqrt{5}$

25. $2^{a-2} = \frac{\sqrt{2}}{16}$

26. $\left(\frac{1}{2}\right)^{-a} = 8$

27. $\frac{2^{a-2}}{4^{1-a}} = 2\sqrt{2}$

28. $x^{a-2} = 1$

29. $2^a \times 4^{a-1} \times 8 = 4$

30. $9^a \times (3^2)^4 = 27$

31. $2^a + 7 \times 2^a = 256$

32. $3\sqrt{2^a} = 6 \times 2^2$

33. $\frac{3^{a-2}}{27^a} = \frac{1}{9^{2a}}$

34. $\frac{1}{25^{a-2}} = \frac{5}{\sqrt{5}}$

35. $6^{2a} = (3^a)^2$

36. $\frac{4 \times 3^a - 21}{3^a + 2} = 3$

Absolute value equations

Examples:

Definition: Case 1: $|x| = x$ if $x > 0$

Case 2: $|x| = -x$ if $x < 0$

① $|x - 2| = 3$

Case 1: $x - 2 = 3$

$$x = 5$$

Case 2: $-(x - 2) = 3$

$$x - 2 = -3$$

$$x = -1$$

② $|2x + 5| = 3x + 9$

Case 1: $2x + 5 = 3x + 9$

$$x = -4$$

Case 2: $-(2x + 5) = 3x + 9$

$$-2x - 5 = 3x + 9$$

$$5x = -14$$

$$x = -2\frac{4}{5}$$

Checking solutions for Example 2 above:

If $x = -4$, LHS = $|2 \times (-4) + 5|$ RHS = $3 \times -4 + 9$

$$= |-8 + 5|$$

$$= -3$$

$$= |-3|$$

$$\neq \text{LHS}$$

$$= 3$$

$\therefore x = -4$ is NOT a solution.

If $x = -2\frac{4}{5}$,

LHS = $|2 \times (-2\frac{4}{5}) + 5|$

$$= |-5\frac{3}{5} + 5|$$

$$= |-1\frac{3}{5}|$$

$$= 1\frac{3}{5}$$

RHS = $3 \times -2\frac{4}{5} + 9$

$$= -8\frac{2}{5} + 9$$

$$= 1\frac{3}{5}$$

$\therefore x = -2\frac{4}{5}$ is a solution, and is the only solution.

Note: Particularly with questions like Example 2, check your solutions.

Solve for x :

1. $|x| = 4$

2. $|3x| = 6$

3. $|x - 4| = 5$

4. $|x + 6| = 2$

5. $|x + 8| = 0$

6. $|5 - x| = 7$

7. $|2x + 3| = 13$

8. $|3x - 2| = 7$

9. $|5x + 8| = 2$

10. $|\frac{2x - 1}{3}| = 5$

11. $|\frac{3x + 4}{7}| = 2$

12. $|x^2| = 8$

13. $|3x + 2| = |2x + 8|$

14. $|7x - 3| = |4x - 5|$

15. $|4x + 8| = 0$

16. $|\frac{5x - 4}{3}| = |\frac{2x + 7}{4}|$

17. $|\frac{3x + 1}{2} + \frac{x - 5}{3}| = 4$

18. $|\frac{2x - 4}{3} - \frac{x - 2}{5}| = \frac{1}{2}$

19. $4|x - 2| = 2|x + 3|$

20. $5|x + 3| - 2|x + 1| = 4$

21. $3|2x + 1| = 4 - |x + 2|$

22. $|3x + 1| = 2x + 4$

23. $|x - 3| = 2x + 1$

24. $|5 - x| = x + 7$

25. $|2x - 3| = 4x + 2$

26. $|3x - 5| = 3x - 5$

27. $|x + 1| + |x - 1| = 6$

28. $|3x + 2| + |x - 4| = 5$

29. $|2x + 3| - |3x + 4| = 7$

30. $|x + 1| + |x + 2| + |x + 3| = 12$

Inequations

Examples: ① $3x - 4 > 5$

$$3x > 9$$

$$x > 3$$

② $4 - 3x > 10$

$$-3x > 6$$

$$x < -2$$

Note: When multiplying or dividing by a negative number, reverse the inequality sign.

Solve for x

1. $4x < 20$

2. $5x \geq -10$

3. $-5x \geq 10$

4. $3x + 2 < 8$

5. $4x - 5 > -1$

6. $7 - 2x < 15$

7. $5 < 3 + 2x$

8. $3(4 - 2x) \geq 6$

9. $4(2x - 1) < 3(x - 5)$

10. $2(x - 4) - (3x + 1) > -2$

11. $20 < 4 - 8x$

12. $\frac{x+3}{2} < -5$

13. $\frac{1-3x}{4} \geq 0$

14. $\frac{x}{5} > \frac{2}{3}$

15. $\frac{3x}{7} + 5 \leq 2$

16. $5 - \frac{x}{3} < -\frac{1}{2}$

17. $2(x - 5) + 3 > 5(3 - x)$

18. $\frac{3x}{-4} < 6$

19. $\frac{2x+3}{5} < \frac{x-2}{3}$

20. $\frac{5-3x}{-4} \geq \frac{x-3}{2}$

21. $\frac{x+1}{3} + \frac{x-2}{4} < 1$

22. $\frac{3x-4}{5} - \frac{x-2}{3} \geq -2$

23. $\frac{4-x}{2} + \frac{3-2x}{5} > 1$

24. $\frac{x}{4} \leq \frac{x}{6} + \frac{1}{3}$

25. $\frac{5-2x}{3} + 4 \leq \frac{1-x}{2}$

26. $\frac{2(x-4)}{3} < \frac{3(5-2x)}{4}$

27. $-5 < 3x + 1 < 7$

28. $\frac{-4x}{-7} > \frac{-3}{2}$

29. $\frac{4(x-3)}{5} - \frac{2(1-4x)}{3} > 2$

30. $3 < 1 - 2x < 7$

31. $\frac{2(x+4)}{3} - \frac{3(1-x)}{2} \geq \frac{3(x-2)}{4}$

Absolute value inequations

Examples: ① $|x - 2| < 3$

$$\text{Case 1: } x - 2 < 3$$

$$x < 5$$

$$\text{Case 2: } -(x - 2) < 3$$

$$-x + 2 < 3$$

$$-x < 1$$

$$x > -1$$

$$\therefore \text{ solution is } -1 < x < 5$$

② $\frac{2}{|x + 3|} < 5$

$$\therefore 2 < 5|x + 3|$$

$$\text{Case 1: } 2 < 5(x + 3)$$

$$2 < 5x + 15$$

$$-13 < 5x$$

$$-2\frac{3}{5} < x$$

$$x > -2\frac{3}{5}$$

$$\text{Case 2: } 2 < -5(x + 3)$$

$$2 < -5x - 15$$

$$17 < -5x$$

$$-3\frac{2}{5} > x$$

$$x < -3\frac{2}{5}$$

$$\therefore \text{ solution is } x < -3\frac{2}{5} \text{ OR } x > -2\frac{3}{5}$$

Solve for x :

1. $|x + 11| < 4$

2. $|2x - 3| \leq 5$

3. $|x - 6| > 6$

4. $|3x + 2| > 4$

5. $|9x - 4| \leq 5$

6. $|x| > 4$

7. $\left|\frac{x}{2}\right| \leq 3$

8. $\left|\frac{2x}{7}\right| > 4$

9. $|1 - 3x| < 4$

10. $|5 + 3x| \geq 0$

11. $\left|\frac{x+1}{4}\right| > 1$

12. $\left|\frac{x-2}{3}\right| \leq 2$

13. $\left|\frac{2x-3}{5}\right| < 2$

14. $\left|\frac{1-3x}{5}\right| > 1$

15. $\left|\frac{4+3x}{2}\right| > 2$

16. $\left|\frac{x-2}{4}\right| \leq \frac{1}{2}$

17. $\left|\frac{2x-1}{3}\right| < \frac{3}{4}$

18. $\left|\frac{x+2}{3}\right| - 5 < 0$

19. $\left|\frac{4-3x}{5}\right| \geq -\frac{3}{4}$

20. $\frac{3}{|x-2|} < 2$

21. $\frac{4}{|2x+1|} \geq 2$

22. $\frac{5}{|3-x|} \leq 4$

23. $\frac{2}{|x-1|} < \frac{3}{4}$

24. $|2x+3| < |x-1|$

25. $\frac{2}{|x-1|} > \frac{3}{|x+2|}$

Inequations with the unknown in the denominator

Example: $\frac{3}{x-1} < 2$

Method 1

Case 1: $3 < 2x - 2$

if $x - 1 > 0$

$$5 < 2x$$

$$x > 1$$

$$2\frac{1}{2} < x$$

$$x > 2\frac{1}{2}$$

Solution for Case 1 is $x > 2\frac{1}{2}$.

Case 2: $3 > 2x - 2$

if $x - 1 < 0$

$$5 > 2x$$

$$x < 1$$

$$2\frac{1}{2} > x$$

$$x < 2\frac{1}{2}$$

Solution for Case 2 is $x < 1$.

Total solution is $x < 1$ OR $x > 2\frac{1}{2}$.

Method 2

Multiply both sides by $(x-1)^2$ (POSITIVE)

$$\frac{3}{x-1} \times (x-1)^2 < 2 \times (x-1)^2$$

$$3(x-1) < 2(x^2 - 2x + 1)$$

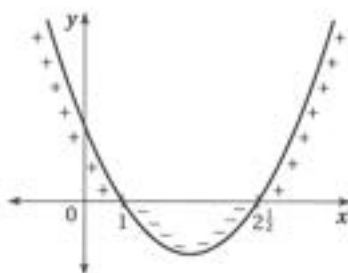
$$3x - 3 < 2x^2 - 4x + 2$$

$$0 < 2x^2 - 7x + 5$$

$$2x^2 - 7x + 5 > 0$$

$$(2x-5)(x-1) > 0$$

$$\therefore x > 2\frac{1}{2} \text{ OR } x < 1$$



Solve for x :

1. $\frac{1}{x-2} < 4$

7. $\frac{3}{2-x} \geq 2$

13. $\frac{4}{|x-1|} < 2$

2. $\frac{1}{x} > 2$

8. $\frac{2}{1-x} > -1$

14. $\frac{2x-1}{4x+2} - 3 > 0$

3. $\frac{2}{x+3} > 1$

9. $\frac{x-2}{x+1} > 0$

15. $\frac{2}{x+1} > \frac{1}{x-1}$

4. $\frac{4}{3x-2} \leq 1$

10. $\frac{x+2}{x-3} > 2$

16. $\frac{1}{x-2} \leq \frac{1}{3x}$

5. $\frac{1}{x} < \frac{1}{4}$

11. $\frac{x+1}{x-4} < \frac{2}{5}$

17. $1 < \frac{3x-2}{x+1} < 4$

6. $\frac{2}{x-4} > \frac{1}{5}$

12. $\frac{3-x}{x} > 2$

18. $2 < \frac{x-4}{2x+1} < 3$

Quadratic inequations

Examples: ① Solve for x : $x^2 - 2x - 3 < 0$

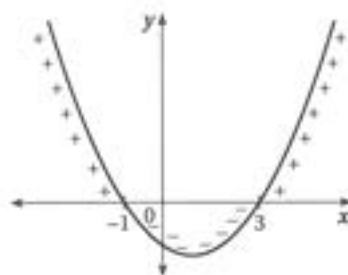
$$(x-3)(x+1) < 0$$

$$\text{If } y = (x-3)(x+1)$$

we require the values which make $y < 0$

ie y negative

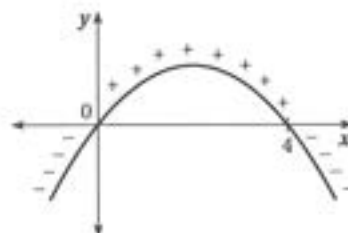
$$\text{ie } -1 < x < 3$$



② Solve for x : $4x - x^2 < 0$

$$x(4-x) < 0$$

$$\therefore x < 0 \text{ and } x > 4$$



Solve for x :

1. $x^2 - 3x - 4 < 0$

12. $2x^2 < 98$

23. $\frac{x^2}{2} < x + \frac{15}{2}$

2. $x^2 - x - 6 > 0$

13. $2x^2 - 4x - 30 < 0$

24. $6x^2 - 25x + 14 > 0$

3. $(x-2)(x+4) < 0$

14. $3x^2 + 36x + 60 > 0$

25. $6x^2 + 39x + 18 < 0$

4. $(x+1)(x+5) > 0$

15. $(3x-5)(x+4) > 0$

26. $10 + 3x - x^2 > 0$

5. $x^2 - 5x - 14 \geq 0$

16. $2x^2 - 50 < 0$

27. $12 + x < 6x^2$

6. $x^2 + 6x < 0$

17. $1 < x^2$

28. $x^2 + 4x + 11 < 0$

7. $x^2 - x - 72 < 0$

18. $(2x-3)^2 > 0$

29. $2x^2 - 3x + 5 > 0$

8. $36 - 13x + x^2 \leq 0$

19. $(2x+3)(5x-2) < 0$

30. $\frac{x^2}{5} < \frac{3x}{2}$

9. $x(x-1) < 12$

20. $2x^2 - 13x - 7 < 0$

31. $6x^2 + 10x + 2 > \frac{1}{2}$

10. $x^2 - 25 > 0$

21. $5x^2 - 22x + 21 > 0$

32. $147 - 3x^2 < 0$

11. $x^2 < 36$

22. $12x^2 - 27 < 0$

33. $x^2 > 27$

Simultaneous equations – Elimination method

Examples: ① Solve for x and y :

$$3x - 2y = 8 \quad \text{_____ (1)}$$

$$x + 2y = 0 \quad \text{_____ (2)}$$

$$(1) + (2): \quad 4x = 8$$

$$x = 2$$

Substitute $x = 2$ into (2)

$$2 + 2y = 0$$

$$2y = -2$$

$$y = -1$$

$\therefore$ Solutions are $x = 2, y = -1$

② $5x + 3y = 19 \quad \text{_____ (1)}$

$$2x - 4y = -8 \quad \text{_____ (2)}$$

$$(1) \times 2: \quad 10x + 6y = 38 \quad \text{_____ (3)}$$

$$(2) \times 5: \quad 10x - 20y = -40 \quad \text{_____ (4)}$$

$$(3) - (4): \quad 26y = 78$$

$$y = 3$$

Substitute $y = 3$ into (1)

$$5x + 9 = 19$$

$$5x = 10$$

$$x = 2$$

$\therefore$ Solutions are $x = 2, y = 3$

Solve the following simultaneous equations:

1. $4x + 2y = 6$

$$x + 2y = 3$$

2. $5x - 3y = 8$

$$2x + 3y = -1$$

3. $3x - y = 4$

$$x - y = 0$$

4. $7x + 3y = 13$

$$3x + 3y = -3$$

5. $5a + 2b = 16$

$$2a - b = 1$$

6. $3a - 4b = 1$

$$a + b = -2$$

7. $a + 2b = 9$

$$2a - 4b = 2$$

8. $6x - 5y = -2$

$$2x + y = 2$$

9. $x + y = 1$

$$x - y = -11$$

10. $2c + 3d = 19$

$$4c - d = 3$$

11. $2f + 3g = 5$

$$3f - 2g = 1$$

12. $5x + 4y = 2$

$$3x - 3y = 12$$

13. $8x + 5y = 19$

$$10x - 2y = -1$$

14. $6a + 4b = 3$

$$9a + 12b = 6$$

15. $2a - 3b = 4$

$$6a + b = 2$$

16. $2c - d = 3$

$$c - 2d = 3$$

17. $4x + 3y - 7 = 0$

$$2x - y - 1 = 0$$

18. $5a - 3b - 14 = 0$

$$2a - 4b - 14 = 0$$

19. $\frac{x}{3} + \frac{y}{2} = \frac{5}{6}$

$$\frac{x}{4} - \frac{y}{3} = \frac{-1}{12}$$

20. $\frac{x}{5} - \frac{y}{2} = \frac{4}{15}$

$$\frac{x}{3} - \frac{y}{4} = \frac{1}{4}$$

Simultaneous equations – Substitution method

Examples: ① $y = 3x$ _____ (1)

$$2x + y = 10 \quad \text{_____ (2)}$$

Substitute (1) in (2)

$$2x + 3x = 10$$

$$5x = 10$$

$$x = 2$$

Substitute $x = 2$ into (1)

$$y = 3 \times 2$$

$$y = 6$$

$\therefore$ Solutions are $x = 2, y = 6$

② $y = x^2 - 3x + 4$ _____ (1)

$$y = 2x - 2 \quad \text{_____ (2)}$$

Substitute (2) in (1)

$$2x - 2 = x^2 - 3x + 4$$

$$x^2 - 5x + 6 = 0$$

$$(x - 3)(x - 2) = 0$$

$$x = 2, 3$$

Substitute $x = 2$ in (2)

$$y = 2 \times 2 - 2$$

$$y = 2$$

Substitute $x = 3$ in (2)

$$y = 2 \times 3 - 2$$

$$y = 4$$

$\therefore$ Solutions are $x = 2, y = 2$ and $x = 3, y = 4$

Solve the following simultaneous equations:

1. $y = 2x$

$$3x + y = 5$$

2. $y = 4x$

$$3x - y = -2$$

3. $y = x$

$$x + y = -2$$

4. $y = x + 4$

$$3x + y = 0$$

5. $y = x - 3$

$$3x + 2y = 4$$

6. $y = x - 3$

$$3x - y = -1$$

7. $y = 2x - 3$

$$4x + 2y = 10$$

8. $y = 4 - x$

$$x - 2y = -11$$

9. $y = 2x + 1$

$$3y - 2x = 4$$

10. $y = \frac{x}{2}$

$$2x - 3y = 4$$

11. $y = 3x - 12$

$$y = x^2 - 4x$$

12. $y = 2x + 1$

$$y = x^2 - 3x - 5$$

13. $y = 2 - 3x$

$$y = x^2 + 5x + 14$$

14. $y = x^2 - 42$

$$y = x$$

15. $y = x + 3$

$$y = 2x^2 - 4x + 5$$

16. $y = x - 3$

$$xy = 4$$

17. $3x + y = 7$

$$xy = 2$$

18. $x^2 - y^2 = 5$

$$x + y = 1$$

19. $x^2 + y^2 = 13$

$$2x - y + 1 = 0$$

20. $y = 5$

$$x^2 + y^2 = 16$$

Equations review

Solve for x :

1. $3x - 4 = 8$

2. $7 - 3x = 2$

3. $3 - x = 3$

4. $\frac{8x}{3} = 2$

5. $\frac{2x}{3} + 5 = 7$

6. $3 - \frac{2x}{5} = 1$

7. $\frac{3x + 12}{2} = 3$

8. $-2(1 + 3x) = -5$

9. $4 + 3x = 5x - 2$

10. $3(1 - x) - (2 - x) = 4$

11. $\frac{5x}{4} = \frac{3}{5}$

12. $\frac{3}{2x} = \frac{4}{5}$

13. $\frac{3x - 2}{5} + \frac{x + 3}{4} = 2$

14. $7 - \frac{3x - 2}{4} = \frac{1 - 2x}{8}$

15. $8^{x-3} = 16$

16. $a^{2x+3} = 1$

17. $2 \times 16^{x-3} = 32$

18. $x^2 - 6x - 7 = 0$

19. $2x^2 - 9x = -9$

20. $x^2 = \frac{x}{2}$

21. $3x^2 - 4x - 2 = 0$

22. $|3x - 2| = 4$

23. $\frac{7}{|3x - 2|} = 4$

24. $|3x + 1| = x - 5$

25. $|x| = 1 - 2x$

26. $5x + 6 > 1$

27. $\frac{7 - 3x}{4} < -2$

28. $\left| \frac{1 - 2x}{5} \right| \geq 3$

29. $\frac{3}{|4x + 1|} < 2$

30. $\frac{5}{x} < 1$

31. $\frac{3}{2x + 1} \geq 1$

32. $x^2 < 25$

33. $x^2 - 2x - 63 < 0$

34. $6 + 5x - x^2 < 0$

Solve for x and y :

35. $3x + 4y = 11$
 $x - 4y = -7$

36. $5x - 3y = 4$
 $3x - 2y = 3$

37. $y = 2x$
 $y = 10 - 3x$

38. $2x - y + 3 = 0$
 $3x + 4y - 6\frac{1}{2} = 0$

39. $y = 2x + 3$
 $y = x^2 - x - 25$

Simplifying surds

Examples: ① Simplify $\sqrt{45}$

$$\begin{aligned}\sqrt{45} &= \sqrt{9 \times 5} \\ &= \sqrt{9} \times \sqrt{5} \\ &= 3 \times \sqrt{5} \\ &= 3\sqrt{5}\end{aligned}$$

② Simplify $3\sqrt{32}$

$$\begin{aligned}3\sqrt{32} &= 3 \times \sqrt{16 \times 2} \\ &= 3 \times 4 \times \sqrt{2} \\ &= 12\sqrt{2}\end{aligned}$$

③ Find x if $2\sqrt{3} = \sqrt{x}$

$$\begin{aligned}\sqrt{4 \times 3} &= \sqrt{x} \\ \sqrt{12} &= \sqrt{x} \\ x &= 12\end{aligned}$$

Simplify:

1. $\sqrt{8}$

2. $\sqrt{27}$

3. $\sqrt{128}$

4. $\sqrt{72}$

5. $\sqrt{99}$

6. $\sqrt{288}$

7. $\sqrt{20}$

8. $\sqrt{54}$

9. $3\sqrt{45}$

10. $5\sqrt{18}$

11. $2\sqrt{50}$

12. $4\sqrt{44}$

13. $4\sqrt{500}$

14. $6\sqrt{147}$

15. $\sqrt{x^2y}$

Find the value of x :

16. $2\sqrt{5} = \sqrt{x}$

17. $3\sqrt{2} = \sqrt{x}$

18. $5\sqrt{3} = \sqrt{x}$

19. $4\sqrt{7} = \sqrt{x}$

20. $\sqrt{x} = 3\sqrt{11}$

21. $3\sqrt{x} = \sqrt{27}$

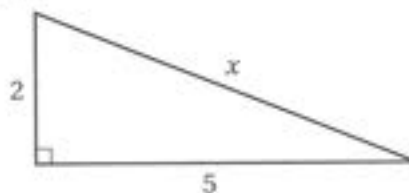
22. $5\sqrt{x} = \sqrt{50}$

23. $2\sqrt{x} = 6$

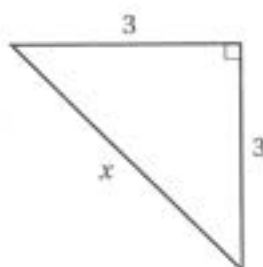
24. $5\sqrt{x} = 1$

Find the *exact* value of x :

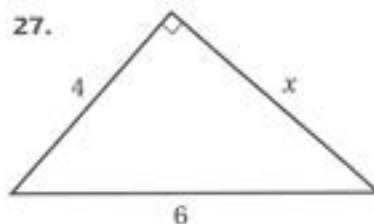
25.



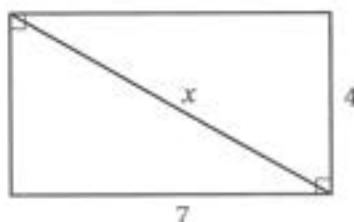
26.



27.



28.



29. Arrange in descending

order: $2\sqrt{15}$, $2\sqrt{3}$, $3\sqrt{2}$, $\sqrt{16}$

30. Arrange in ascending

order: $2\sqrt{13}$, $4\sqrt{3}$, $5\sqrt{2}$, 7 , $\sqrt{51}$

Simplify:

31. $\sqrt[3]{54}$

32. $\sqrt[3]{x^6y^9}$

33. $5\sqrt[3]{250}$

Addition and subtraction of surds

Examples: ① $\sqrt{8} + \sqrt{50} = \sqrt{4 \times 2} + \sqrt{25 \times 2}$ ② $2\sqrt{3} - 4\sqrt{48} = 2\sqrt{3} - 4\sqrt{16 \times 3}$

$$= 2\sqrt{2} + 5\sqrt{2} = 2\sqrt{3} - 4 \times 4\sqrt{3}$$

$$= 7\sqrt{2} = 2\sqrt{3} - 16\sqrt{3}$$

$$= -14\sqrt{3}$$

③ $\sqrt{20} + 3\sqrt{45} - \sqrt{27} = \sqrt{4 \times 5} + 3\sqrt{9 \times 5} - \sqrt{9 \times 3}$

$$= 2\sqrt{5} + 9\sqrt{5} - 3\sqrt{3}$$

$$= 11\sqrt{5} - 3\sqrt{3}$$

Simplify:

1. $\sqrt{5} + 3\sqrt{5} - 2\sqrt{5}$

2. $2\sqrt{5} - \sqrt{5} - \sqrt{5}$

3. $\sqrt{5} + \sqrt{3} + \sqrt{2}$

4. $2\sqrt{5} - 3\sqrt{2} + \sqrt{5} - 4\sqrt{2}$

5. $\sqrt{8} + \sqrt{2}$

6. $3\sqrt{27} - 5\sqrt{3}$

7. $\sqrt{50} + \sqrt{75}$

8. $3\sqrt{2} - \sqrt{98} + 2\sqrt{8}$

9. $7 + \sqrt{20} - 2\sqrt{45}$

10. $\sqrt{24} - 2\sqrt{44} + \sqrt{54} - 3\sqrt{99}$

11. $\sqrt{3} + 3\sqrt{75} - 2\sqrt{48}$

12. $-5 - \sqrt{3} - \sqrt{27} - 3$

13. $\sqrt{28} - 4\sqrt{8} + 2\sqrt{63}$

14. $\sqrt{16} + \sqrt{32} + \sqrt{48}$

15. $2\sqrt{125} - 7\sqrt{5} + \sqrt{50} - 2\sqrt{5}$

16. $\sqrt{33} + \sqrt{21} - \sqrt{132} + \sqrt{189}$

17. $2\sqrt{80} - 4\sqrt{12} + 2\sqrt{20} + \sqrt{3}$

18. $-\sqrt{2} - \sqrt{3} - 2\sqrt{8} - 3\sqrt{27}$

19. $\sqrt{98} + \sqrt{99} + \sqrt{32} - \sqrt{176}$

20. $3\sqrt{72} - 2\sqrt{48} - 4\sqrt{162} + 5\sqrt{75}$

21. $\sqrt{25} - 2\sqrt{50} - \sqrt{49} + 3\sqrt{98}$

22. $\sqrt{a} + \sqrt{9a}$

23. $\sqrt{363} - 4\sqrt{243} + 2\sqrt{192}$

24. $2\sqrt{63} - 4\sqrt{28} + \sqrt{112}$

25. $2\sqrt{3} - 2 + 3\sqrt{2} - 3$

26. $3\sqrt{175} + \sqrt{200} - 2\sqrt{63} - \sqrt{32}$

27. $\sqrt{16B} - \sqrt{25B} + \sqrt{81B}$

28. $3\sqrt{90} - \sqrt{400} + 2\sqrt{160} + \sqrt{900}$

29. $\sqrt{x^3} + x\sqrt{x} - \sqrt{y^2x}$

30. $2\sqrt{24} - 5\sqrt{54} + \frac{1}{2}\sqrt{216} - \frac{\sqrt{294}}{7}$

Multiplication of surds

Examples: ① $\sqrt{3} \times \sqrt{5} = \sqrt{15}$

② $2\sqrt{3} \times 4\sqrt{5} = 8\sqrt{15}$

③ $\sqrt{3} \times \sqrt{3} = \sqrt{9}$
 $= 3$

④ $5(\sqrt{3} + 2) + 2(\sqrt{3} - 4) = 5\sqrt{3} + 10 + 2\sqrt{3} - 8$
 $= 7\sqrt{3} + 2$

⑤ $2\sqrt{98} \times 5\sqrt{32} = 2 \times 7\sqrt{2} \times 5 \times 4\sqrt{2}$
 $= 280\sqrt{4}$
 $= 280 \times 2$
 $= 560$

Simplify:

1. $\sqrt{2} \times \sqrt{3}$

2. $\sqrt{5} \times \sqrt{5}$

3. $\sqrt{2} \times 3\sqrt{2}$

4. $\sqrt{8} \times \sqrt{6}$

5. $\sqrt{9} \times 3\sqrt{3}$

6. $2\sqrt{2} \times 5\sqrt{2}$

7. $4\sqrt{6} \times 2\sqrt{3}$

8. $\sqrt{3} \times \sqrt{3} \times \sqrt{3}$

9. $\sqrt{10} \times \sqrt{5} \times \sqrt{2}$

10. $(\sqrt{7})^2$

11. $(2\sqrt{3})^2$

12. $(3\sqrt{6})^2$

13. $\sqrt{8} \times \sqrt{18}$

14. $\sqrt{3} \times \sqrt{2} \times \sqrt{3}$

15. $(\sqrt{5})^2 \times 2\sqrt{2}$

16. $(2\sqrt{3})^3$

Expand and simplify:

17. $5(\sqrt{3} + 2)$

18. $7(4 - 2\sqrt{7})$

19. $\sqrt{3}(2\sqrt{3} + 4)$

20. $2\sqrt{2}(3\sqrt{2} - 2)$

21. $4(\sqrt{2} + 3) + 2(\sqrt{2} + 1)$

22. $5(2\sqrt{3} + \sqrt{5}) + 3(\sqrt{3} + 2\sqrt{5})$

23. $3(\sqrt{2} - 3) + 4(\sqrt{8} - 2)$

24. $7(2\sqrt{10} - 5) - 3(\sqrt{90} + 2)$

25. $\sqrt{2}(1 - 3\sqrt{2}) + \sqrt{3}(2 + 4\sqrt{3})$

26. $4(2\sqrt{3} - 3\sqrt{2}) - 5(\sqrt{27} - 2\sqrt{18})$

27. $16 - 6(2\sqrt{3} - 3)$

28. $\sqrt{2}(2\sqrt{50} - \sqrt{45}) - \sqrt{10}(4 + \sqrt{50})$

29. $-(\sqrt{3} - 2\sqrt{2}) - (\sqrt{32} + \sqrt{243})$

30. $2\sqrt{2}(\sqrt{2} - 4) - 3(\sqrt{2} + 1) - \sqrt{8}(2\sqrt{2} + 1)$

Binomial products involving surds

Examples: ① $(\sqrt{3} + 2)(\sqrt{3} + 5) = 3 + 5\sqrt{3} + 2\sqrt{3} + 10$
 $= 13 + 7\sqrt{3}$

② $(2\sqrt{3} - 5)^2 = 4\sqrt{9} - 20\sqrt{3} + 25$
 $= 12 - 20\sqrt{3} + 25$
 $= 37 - 20\sqrt{3}$

③ $(2\sqrt{3} + \sqrt{5})(2\sqrt{3} - \sqrt{5}) = 4\sqrt{9} - 5$
 $= 12 - 5$
 $= 7$

Expand and simplify:

1. $(\sqrt{5} + 2)(\sqrt{3} + 7)$

16. $(3\sqrt{2} + 2\sqrt{3})^2$

2. $(\sqrt{3} + 1)(\sqrt{3} + 4)$

17. $(\sqrt{5} - 2)(\sqrt{5} + 2)$

3. $(2\sqrt{3} + 4)(3\sqrt{3} + 6)$

18. $(2\sqrt{3} + 3)(2\sqrt{3} - 3)$

4. $(\sqrt{5} - 2)(\sqrt{5} + 6)$

19. $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b})$

5. $(\sqrt{3} + \sqrt{2})(5 - \sqrt{2})$

20. $(\sqrt{a} + \sqrt{b})^2$

6. $(2\sqrt{7} - \sqrt{5})(3\sqrt{7} + 2\sqrt{5})$

21. $(\sqrt{a} - 2\sqrt{b})^2$

7. $(4\sqrt{3} - 5)(2\sqrt{3} - 2)$

22. $(\sqrt{3} + 1)^2 + (2\sqrt{3} + 5)^2$

8. $(\sqrt{32} - 2\sqrt{27})(\sqrt{50} + \sqrt{12})$

23. $(2\sqrt{5} + \sqrt{2})^2 - (3\sqrt{2} - 2\sqrt{5})^2$

9. $(\sqrt{a} + \sqrt{b})(\sqrt{a} + 2\sqrt{b})$

24. $(\sqrt{5} - 2)^2 - (\sqrt{5} + 1)(2\sqrt{5} + 3)$

10. $(\sqrt{7} + \sqrt{5})(\sqrt{7} + \sqrt{5})$

25. $5\sqrt{2} - (\sqrt{2} + 1)^2$

11. $(\sqrt{3} + \sqrt{2})^2$

26. $(2\sqrt{2} + 3)(4 - 3\sqrt{2}) + (2\sqrt{2} + 1)^2$

12. $(2\sqrt{3} - 1)^2$

27. $(5\sqrt{3} + 4)(2\sqrt{27} - \sqrt{2}) - (\sqrt{2} + \sqrt{3})(2\sqrt{12} - 3\sqrt{8})$

13. $(3\sqrt{5} + 2)^2$

28. $(\sqrt{a} - \sqrt{b})^2 - (\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b})$

14. $(4 - 2\sqrt{7})^2$

29. $(2\sqrt{3} + 1)^3$

15. $(\sqrt{5} - 2\sqrt{6})^2$

30. $(2\sqrt{3} + 3\sqrt{2})^2 - (2\sqrt{3} - 3\sqrt{2})^2 - (2\sqrt{3} + 3\sqrt{2})(2\sqrt{3} - 3\sqrt{2})$

Division of surds

Examples: ① $\frac{\sqrt{6}}{\sqrt{2}} = \sqrt{\frac{6}{2}}$
 $= \sqrt{3}$

② $\frac{\sqrt{8}}{2} = \frac{2\sqrt{2}}{2}$
 $= \sqrt{2}$

③ $\frac{\sqrt{80}}{\sqrt{20}} = \frac{\sqrt{16 \times 5}}{\sqrt{4 \times 5}}$ OR $\frac{\sqrt{80}}{\sqrt{20}} = \sqrt{\frac{80}{20}}$
 $= \frac{4\sqrt{5}}{2\sqrt{5}}$ $= \sqrt{4}$
 $= 2$ $= 2$

Simplify:

1. $\frac{\sqrt{10}}{\sqrt{2}}$

12. $\frac{\sqrt{48}}{\sqrt{16}}$

23. $\frac{5\sqrt{2} \times 4\sqrt{2}}{\sqrt{8}}$

2. $\frac{\sqrt{15}}{\sqrt{5}}$

13. $\frac{5\sqrt{2}}{\sqrt{50}}$

24. $\frac{\sqrt{5} \times 2\sqrt{6}}{\sqrt{3} \times \sqrt{10}}$

3. $\frac{\sqrt{20}}{\sqrt{5}}$

14. $\sqrt{243} \div \sqrt{108}$

25. $\frac{3\sqrt{2} \times 2\sqrt{6}}{4\sqrt{3}}$

4. $\frac{\sqrt{32}}{\sqrt{18}}$

15. $\sqrt{200} \div \sqrt{2}$

26. $\frac{(2\sqrt{3})^2}{(3\sqrt{2})^2}$

5. $5\sqrt{2} \div 5$

16. $\sqrt{\frac{24}{32}}$

27. $\sqrt{15} \div \sqrt{5} \times \sqrt{3}$

6. $5\sqrt{2} \div \sqrt{2}$

17. $4\sqrt{98} \div 7$

28. $\sqrt{24} \times 2\sqrt{2} \div \sqrt{2}$

7. $\sqrt{8} \div 2\sqrt{2}$

18. $\frac{3\sqrt{242}}{\sqrt{2}}$

29. $\left(\frac{\sqrt{3}}{2}\right)^2 \div \left(\frac{\sqrt{3}}{5}\right)^2$

8. $\frac{\sqrt{6}}{\sqrt{24}}$

19. $\sqrt{98} \div 2\sqrt{2}$

30. $\frac{3\sqrt{32}}{4\sqrt{48}} \times \frac{2\sqrt{27}}{4\sqrt{50}}$

9. $\frac{3\sqrt{5}}{\sqrt{125}}$

20. $\frac{\sqrt{20}}{\sqrt{10} \times 2\sqrt{2}}$

31. $\frac{2\sqrt{28}}{3\sqrt{40}} \div \frac{\sqrt{112}}{2\sqrt{90}}$

10. $6\sqrt{3} \div \sqrt{27}$

21. $\frac{2\sqrt{3} \times 3\sqrt{2}}{\sqrt{6}}$

32. $\frac{(\sqrt{a^2})^3}{(\sqrt[3]{a^3})^2}$

11. $\frac{\sqrt{8}}{8}$

22. $\frac{(2\sqrt{3})^2}{3}$

33. $\frac{\sqrt{27}}{3\sqrt{3}} \div (2\sqrt{2} \div \sqrt{2} \times 2)$

Rationalising denominators

Examples: Express with a rational denominator in its simplest form:

$$\begin{aligned} \textcircled{1} \quad \frac{7}{\sqrt{3}} &= \frac{7}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \\ &= \frac{7\sqrt{3}}{3} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad \frac{5}{3\sqrt{2}} &= \frac{5}{3\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\ &= \frac{5\sqrt{2}}{6} \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad \frac{5}{3-\sqrt{2}} &= \frac{5}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}} \\ &= \frac{5(3+\sqrt{2})}{9-2} \\ &= \frac{5(3+\sqrt{2})}{7} \end{aligned}$$

Express with a rational denominator in its simplest form:

1. $\frac{5}{\sqrt{2}}$

2. $\frac{8}{\sqrt{3}}$

3. $\frac{\sqrt{2}}{\sqrt{7}}$

4. $\frac{2}{\sqrt{2}}$

5. $\frac{2\sqrt{3}}{\sqrt{6}}$

6. $\frac{3\sqrt{5}}{\sqrt{8}}$

7. $\frac{4}{\sqrt{12}}$

8. $\frac{3}{2\sqrt{2}}$

9. $\frac{9}{2\sqrt{3}}$

10. $\frac{5\sqrt{2}}{2\sqrt{5}}$

11. $\frac{6}{4\sqrt{2}}$

12. $\frac{5}{2\sqrt{5}}$

13. $\frac{2}{\sqrt{5}+\sqrt{3}}$

14. $\frac{3}{3-\sqrt{6}}$

15. $\frac{2}{2\sqrt{2}+3}$

16. $\frac{\sqrt{3}}{\sqrt{3}-1}$

17. $\frac{5\sqrt{2}}{\sqrt{5}+\sqrt{3}}$

18. $\frac{3\sqrt{3}}{\sqrt{5}-\sqrt{2}}$

19. $\frac{27}{6\sqrt{2}-3\sqrt{6}}$

20. $\frac{\sqrt{2}-1}{\sqrt{2}+1}$

21. $\frac{\sqrt{7}-\sqrt{5}}{\sqrt{7}+\sqrt{5}}$

22. $\frac{2\sqrt{3}-3}{3\sqrt{2}-\sqrt{6}}$

23. If $x = \sqrt{3} + 1$,
evaluate $x - \frac{1}{x}$.

24. If $x = 2\sqrt{3} - \sqrt{2}$,
evaluate $x - \frac{1}{x}$.

25. If $c = 3\sqrt{2} + \sqrt{3}$,
evaluate $\frac{1}{c^2}$.

26. If $\frac{1}{2\sqrt{5} + \sqrt{2}} =$
 $a\sqrt{5} + b\sqrt{2}$,
evaluate a and b .

27. If $\frac{\sqrt{3}-1}{\sqrt{3}+1} = a + b\sqrt{3}$,
evaluate a and b .

28. $\frac{1}{\sqrt{5}-\sqrt{2}} + \frac{1}{\sqrt{5}+\sqrt{2}}$

29. $\frac{\sqrt{7}+2}{\sqrt{7}-2} - \frac{\sqrt{7}-2}{\sqrt{7}+2}$

30. $\frac{\sqrt{5}+\sqrt{2}}{\sqrt{5}-\sqrt{2}} \div \frac{2\sqrt{5}-3\sqrt{2}}{\sqrt{5}+2\sqrt{2}}$

Surds review

Simplify:

1. $\sqrt{50}$

2. $3\sqrt{63}$

3. Find x if $2\sqrt{x} = 5$.

4. $\sqrt{8} - 2\sqrt{32}$

5. $2\sqrt{12} + 5\sqrt{75} - 3\sqrt{27}$

6. $\sqrt{9} + \sqrt{20} + \sqrt{80} + \sqrt{16}$

7. $2\sqrt{3} - 4\sqrt{24} + \sqrt{48} - 3\sqrt{54}$

8. $2\sqrt{32} - 5\sqrt{72} + \sqrt{144}$

9. $\frac{1}{2}\sqrt{72} - 2\sqrt{125} - \frac{\sqrt{32}}{4} + \sqrt{180}$

10. $2\sqrt{a} + \sqrt{16a} - \sqrt{a^3}$

11. $\sqrt{5} \times \sqrt{6}$

12. $\sqrt{5} \times \sqrt{5} \times \sqrt{5}$

13. $2\sqrt{6} \times 4\sqrt{8}$

14. $2\sqrt{3} \times 2\sqrt{3}$

15. $(-3\sqrt{5})^2$

16. $(3\sqrt{2})^3$

17. $\sqrt{2}(2\sqrt{2} - 2)$

18. $3(\sqrt{2} - 5) - 4(3 - \sqrt{2})$

19. $\sqrt{2}(3 - 4\sqrt{2}) + \sqrt{3}(2\sqrt{3} - 5)$

20. $(2\sqrt{3} + 5)(\sqrt{3} - 2)$

21. $(3\sqrt{7} - 4)(3\sqrt{7} + 4)$

22. $(3\sqrt{2} + 2\sqrt{3})^2$

23. $(2\sqrt{3} - 4\sqrt{6})^2$

24. $(2\sqrt{3} + 4)(2\sqrt{3} - 4) - (4 - 2\sqrt{3})$

25. $(3\sqrt{2} + 2\sqrt{5})^2 - (3\sqrt{2} - 2\sqrt{5})^2$

26. $\frac{\sqrt{12}}{\sqrt{2}}$

27. $\frac{2\sqrt{84}}{6}$

28. $\frac{3\sqrt{6} \times 4\sqrt{8}}{6\sqrt{3}}$

29. $\sqrt{24} \div \sqrt{6} \times \sqrt{8}$

Rationalise the denominators and simplify:

30. $\frac{6}{\sqrt{2}}$

31. $\frac{4}{3\sqrt{2}}$

32. $\frac{4}{3 - \sqrt{2}}$

33. $\frac{2\sqrt{3} + \sqrt{2}}{2\sqrt{3} - \sqrt{2}}$

34. $\frac{2}{\sqrt{6} - \sqrt{2}} - \frac{3}{\sqrt{6} + \sqrt{2}}$

Answers

1. Addition and subtraction

- 1 $3a+4b$ 2 $11x$ 3 $6c+5$ 4 $7x-7$ 5 $4a$ 6 $5a^2$
 7 a^2-a 8 a^3+a^2 9 $7cg$ 10 $4x^2y$ 11 $4a^2c^3-2a^3c^2$
 12 $2a^2$ 13 $7a+7b$ 14 $10a-3b$ 15 $6a-4b$
 16 $-5c-a$ 17 $2a$ 18 $2a^2-5a$ 19 $4a^3+a$
 20 $5a-2b-c$ 21 $-2a^2b+5b^2a$ 22 $8ab-7a-b$
 23 $16gh-2g^2-3h$ 24 $-a^2+b^2$ 25 $ab+ca$
 26 $7a^4+4a^3-4a^2-4a$ 27 $a^2+4a-15$ 28 $3\pi r^2+2\pi r$
 29 $2a^3+8$ 30 $\frac{y^2}{2}-2x^2+xy^2$

2. Multiplication and division

- 1 ab 2 a^3 3 $8cg$ 4 $6a^2$ 5 $-30c$ 6 $24abc$ 7 $2a$
 8 4 9 $\frac{b}{2}$ 10 $\frac{2a}{b}$ 11 $3a$ 12 a^9 13 1 14 $6a^3b^2$
 15 $60a^2b$ 16 2 17 $40a^6$ 18 a^2 19 $3a^7$ 20 $\frac{c^4}{3}$
 21 $30a^6b^{10}$ 22 $28a^4b^3$ 23 $\frac{2a^2}{b^3}$ 24 $4c^6$ 25 $\frac{a^2b^2}{3}$
 26 30 27 $2a$ 28 $2c^2d^5$ 29 $\frac{3}{2}$ 30 $\frac{b^0}{2}$

3. Substitution

- 1 8 2 7 3 -5 4 32 5 0 6 $\frac{4}{9}$ 7 9 8 -21
 9 -1 10 8 11 8 12 -3 13 $\frac{16}{81}$ 14 0 15 9
 16 *undefined* 17 $\frac{1}{6}$ 18 5 19 $\frac{-3}{4}$ 20 -12 21 $\frac{5}{6}$
 22 $\frac{3}{4}$ 23 576 24 $5\frac{4}{9}$ 25 -33 26 16 27 -2
 28 $\frac{3}{4}$ 29 343 30 $-2\frac{1}{6}$

4. Negative indices

- 1 $\frac{1}{a^2}$ 2 $\frac{2}{a^5}$ 3 $\frac{a^4}{b^2}$ 4 $\frac{7b}{a}$ 5 $\frac{1}{a^2}$ 6 $\frac{2}{a}$ 7 $\frac{1}{a^6}$ 8 $\frac{1}{a}$
 9 $\frac{6}{a^5}$ 10 $\frac{1}{b^2}$ 11 $\frac{2}{c^3}$ 12 12 13 $\frac{8}{c^{12}}$ 14 $\frac{1}{f^{10}}$ 15 c^4
 16 $\frac{1}{g^6}$ 17 $\frac{b^8}{a^6}$ 18 $\frac{c^4}{4b^6}$ 19 8 20 $\frac{a^2}{4}$ 21 $\frac{2}{a^2}$
 22 $\frac{c}{2b^3}$ 23 1 24 $\frac{49}{25a^4}$ 25 $\frac{10}{3c}$ 26 $\frac{b^{14}}{a^2}$ 27 $2\frac{1}{4}$
 28 $\frac{4a^5}{b^7}$ 29 $\frac{108a^3}{b^{14}}$ 30 $\frac{16b^6}{a^5}$

5. Fractional indices

- 1 6 2 2 3 2.14 4 $\frac{1}{2}$ 5 9 6 $\frac{1}{32}$ 7 $\frac{1}{5}$ 8 $\frac{7}{7}$ 9 $\frac{2}{3}$
 10 $1\frac{1}{2}$ 11 $3\frac{3}{8}$ 12 a 13 $\sqrt[4]{a^5}$ 14 $\sqrt[4]{b^5}$ 15 $a^{\frac{1}{12}}$
 16 $\frac{2}{c^{\frac{1}{3}}}$ 17 1 18 c^3 19 $\frac{5}{2a^3}$ 20 $\frac{1}{\sqrt{b^3}}$ 21 $\frac{1}{c^{\frac{1}{3}}}$
 22 $\frac{b^{\frac{1}{3}}}{2}$ 23 $\sqrt[4]{a^7}$ 24 1.25 25 7.25 26 $\sqrt{a^7}\sqrt[3]{b}$
 27 $\frac{2}{\sqrt{b^3}}$ 28 $\frac{2}{a^5}$ 29 $\sqrt[3]{1000}$ 30 $\frac{a^2\sqrt[3]{c^5}}{b}$

6. Distributive law

- 1 $6a+24$ 2 $6a-15$ 3 $12a+8b$ 4 a^2+3a
 5 $21b-6b^3$ 6 $12b^5-8b^3a^2$ 7 $-4a^2+12$ 8 $-1+a$
 9 $-2-4a$ 10 $10a+15b-5c$ 11 $5a^3-15a^2$
 12 $8b^2c-12bc^2$ 13 $4c-7$ 14 $a^2-2ab+b^2$
 15 $15-2a$ 16 $-a-1$ 17 $3a^2-a^3$ 18 $8x+10y$
 19 b^2 20 $3a+b$ 21 $6a+20$ 22 $12c-43$
 23 $2a-10$ 24 $3a^3-7a^2$ 25 $-a^2-7ab+2a^3$
 26 $x-4y$ 27 $5a^2-a$ 28 $5ab-4b^2$ 29 $-3x+5y$
 30 $5a^3-5$

7. Binomial expansions

- 1 a^2+5a+6 2 $b^2+2b-35$ 3 c^2-3c+2
 4 $2a^2+5a-12$ 5 $2a^2+5ab+3b^2$ 6 $-6a^2+14a-4$
 7 $-a^2+2ab-b^2$ 8 a^2+6a+9 9 $4a^2+20a+25$
 10 $a^2-14a+49$ 11 $b^2-4ab+4a^2$ 12 $25c^2+20cg+4g^2$
 13 $9b^4-12ab^2+4a^2$ 14 a^2-4 15 $9a^2-25b^2$
 16 $4c^6-c^4$ 17 $9a^2-24ab+16b^2$ 18 a^4-4a^2
 19 $4a^4$ 20 $2a^2+14a+25$ 21 $2x^2+x-12$ 22 $c+10$
 23 $-9b+22$ 24 $-4a+12$ 25 $9-44a-5a^2$
 26 $2f^3-9f^2-17f-6$ 27 $-7b^4-6b^3+13b^2+3b-5$
 28 $8a^3-12a^2b+6ab^2-b^3$ 29 $8a-8$
 30 $8a^2-12ab+8b^2$

8. Basic algebra and indices review

- 1 $4a$ 2 $5a-5$ 3 $5a-4b$ 4 $6a-4b$ 5 a^2-a
 6 $6a^{10}$ 7 $\frac{b^4}{2}$ 8 $27a^6b^3$ 9 $2a$ 10 3 11 $\frac{6}{a^4}$
 12 $\frac{16}{49}$ 13 $\frac{2b}{3a^4}$ 14 $\frac{2}{a^3}$ 15 $\frac{16c^4}{a^2b^5}$ 16 4 17 $\frac{1}{32}$
 18 $\frac{-1}{8}$ 19 $\frac{4}{9}$ 20 a^5 21 $\frac{1}{a}$ 22 1 23 $\frac{1}{a^8}$ 24 -2
 25 5 26 -4 27 $10a+2$ 28 $5ab-3a^2$ 29 $4a+6a^2$
 30 $9a^2-12ab+4b^2$ 31 $25a^2-9c^2$ 32 $-16a$
 33 $-5a-13$ 34 $3a^2+3ab+7b^2$

9. Common factors

- 1 $5(a+3)$ 2 $8(a-2)$ 3 $2(2a+3b)$ 4 $3(a+2b+4c)$
 5 $b^4(1-b^2)$ 6 $2c(4a+3e)$ 7 $x(a-b+c)$ 8 $2y(y+2)$
 9 $x(x-y)$ 10 $2b(2b^2-3b-4)$ 11 $a(c+d-e)$
 12 $7(5-8y)$ 13 $m(1-m^2)$ 14 $\frac{a}{2}(1+2b)$ 15 $xy(x+y)$
 16 $a^2b^4(a-b^2)$ 17 $-5b(3b+2)$ 18 $a(3a+b+c)$
 19 $5c(c-3)$ 20 $4cd^3(c+2d)$ 21 $(a+b)(3+c)$
 22 $(x+a)(1+x+a)$ 23 $x(3x+y+t)$ 24 $3a^2(3a^2-2+4a^3)$ 25 $5xy^4(3x^2+5y^2)$ 26 $\frac{1}{6}(4a+5b)$ 27 $\sqrt{x}(1+\sqrt{x})$
 28 $(x+y^2)(1+4x+4y)$ 29 $4x^2yz(2y-3)$
 30 $5ab(2a^2b^3-5b^5+6a^2)$

10. Difference of two squares

- 1 $(a-4)(a+4)$ 2 $(b-3)(b+3)$ 3 $(c-11)(c+11)$
 4 $(a-b)(a+b)$ 5 $(1+a)(1-a)$ 6 $(7-x)(7+x)$
 7 $(3a+5)(3a-5)$ 8 $(6b-7c)(6b+7c)$ 9 $(4+5a)(4-5a)$
 10 $5(a+4)(a-4)$ 11 $(ab-c)(ab+c)$ 12 $(1-7m)(1+7m)$
 13 $a(b+a)(b-a)$ 14 $2(2c-b)(2c+b)$ 15 $x(x+1)(x-1)$
 16 $(a-2)(4+a)$ 17 $2(9a-5c)(9a+5c)$ 18 $(a^2-7)(a^2+7)$
 19 $(\frac{x}{a}-\frac{y}{b})(\frac{x}{a}+\frac{y}{b})$ 20 $3d^2(c-4)(c+4)$ 21 $2(a+3)(a-3)$
 22 $(9a-5c)(9a+5c)$ 23 $3(4c+5b)(4c-5b)$
 24 $(ab+cd^2)(ab-cd^2)$ 25 $3(a+c-3)(a+c+3)$
 26 $7(2a+3)$ 27 $5b^2(a+3b)(a-3b)$
 28 $(m-1)(m+1)(m^2+1)$ 29 $4(x+2y+2z)(x-2y-2z)$
 30 $5a^2(b^2+4c^2)(b-2c)(b+2c)$

11. Grouping

- 1 $(a+2)(b+c)$ 2 $(2b-5c)(3a-g)$ 3 $(a+b)(6+c)$
 4 $(a-2)(5b+1)$ 5 $(a+1)(a+b)$ 6 $(2y+3c)(x+2g)$
 7 $2(2a+g)(2b-5c)$ 8 $(b-1)(b-c)$ 9 $(a-4)(a^2+2)$
 10 $(x^2-10)(x-5)$ 11 $(c-3)(c-1)(c+1)$
 12 $(2-a)(3a+b)$ 13 $(c-4)(c^2+2)$ 14 $(ab+6)(a-b)$
 15 $(x-1)(x^3+a)$ 16 $(c-e)(d+e)$ 17 $(3b-2t)(5a+g)$
 18 $(c-d)(a-b)$ 19 $(a+d)(c-7)$ 20 $(10+a)(a-12b)$
 21 $(b^2+c)(a-y)$ 22 $(x+1)(x^2-2)$ 23 $(ab-c)(b+e)$

$$24 \ 3(a-1)(a^2+1) \quad 25 \ (a+b)(a-b+2)$$

$$26 \ (2x+5y)(2x-5y-1) \quad 27 \ (a-b)(1-a-b)$$

$$28 \ (2a-1)\left(\frac{b}{4}+5\right) \quad 29 \ 5(2a-3b)(4b-3a)$$

$$30 \ (x-a)[(x+a)(x^2+a^2)+2]$$

12. Monic trinomials

$$1 \ (a+5)(a+2) \quad 2 \ (b+13)(b+1) \quad 3 \ (x+3)(x+2)$$

$$4 \ (x-8)(x+1) \quad 5 \ (c-4)(c+2) \quad 6 \ (y+6)(y-1)$$

$$7 \ (a-6)(a-2) \quad 8 \ (x+7)(x-9) \quad 9 \ (a+25)(a+4)$$

$$10 \ (b-7)(b+5) \quad 11 \ (x-21)(x+2) \quad 12 \ (x-14)(x-3)$$

$$13 \ (x+7)(x-6) \quad 14 \ (x-15)(x+1) \quad 15 \ (a+9)(a-11)$$

$$16 \ (b-2)(b-16) \quad 17 \ (c-9)(c+8) \quad 18 \ (g+24)(g+6)$$

$$19 \ (x-3)^2 \quad 20 \ (b-4)(b-18) \quad 21 \ 2(b-3)(b+1)$$

$$22 \ 3(a+5)(a+2) \quad 23 \ 3(x+8)(x-1) \quad 24 \ a(b-9)(b-4)$$

$$25 \ 2(x+72)(x+2) \quad 26 \ c(c+4)(c+7) \quad 27 \ (x-y)^2$$

$$28 \ (a-3b)(a+2b) \quad 29 \ 7(c+18b)(c-2b)$$

$$30 \ \frac{a^2}{2}(a-5)(a+4)$$

13. Non-monic trinomials

$$1 \ (2a+1)(a+7) \quad 2 \ (3b+2)(b+5) \quad 3 \ (2a-5)(a-3)$$

$$4 \ (5c+1)(c-8) \quad 5 \ (3a-4)(a+4) \quad 6 \ (5a-2)(a+1)$$

$$7 \ (3x+1)(x+4) \quad 8 \ 5(2a-1)(a+2) \quad 9 \ (2b-1)(b-4)$$

$$10 \ 2(3a-2)(a+5) \quad 11 \ (11c+3)(c+2) \quad 12 \ (7a-5)(a+1)$$

$$13 \ 3(3x-2)(x-5) \quad 14 \ (x+2)(1-x) \quad 15 \ (5x+1)(2x-3)$$

$$16 \ (3a-2)(2a-5) \quad 17 \ (5T+1)(T+1) \quad 18 \ (3a+1)(4-a)$$

$$19 \ (3a+5)^2 \quad 20 \ (6c+1)(3c-2) \quad 21 \ 4(3b-2)(2b-1)$$

$$22 \ (5a+b)(a-7b) \quad 23 \ (3a+4)^2 \quad 24 \ (4x-5)(3x-4)$$

$$25 \ (5x+24)(x-3) \quad 26 \ 6(5p+1)(p-3)$$

$$27 \ (5a+2)(1-2a) \quad 28 \ 4(3x+2y)(x-5y)$$

$$29 \ c(4c+1)(2c-5) \quad 30 \ \frac{1}{2}(7a-6)(a-2)$$

14. Sum and difference of two cubes

$$1 \ (x+1)(x^2-x+1) \quad 2 \ (x-1)(x^2+x+1)$$

$$3 \ (a+3)(a^2-3a+9) \quad 4 \ (a-5)(a^2+5a+25)$$

$$5 \ 2(x-3)(x^2+3x+9) \quad 6 \ (6-b)(36+6b+b^2)$$

$$7 \ (2y+3)(4y^2-6y+9) \quad 8 \ 3(a+5)(a^2-5a+25)$$

$$9 \ (a-4b)(a^2+4ab+16b^2) \quad 10 \ 7(m-1)(m^2+m+1)$$

$$11 \ b(a+c)(a^2-ac+c^2) \quad 12 \ \pi(R-r)(R^2+Rr+r^2)$$

$$13 \ 8(2c+b)(4c^2-2cb+b^2) \quad 14 \ x^2(x+2)(x^2-2x+4)$$

$$15 \ b(3a^2+3ab+b^2) \quad 16 \ (x-y)(x+y)(x^2+x^2y^2+y^2)$$

$$17 \ 2(3a+2)(9a^2-6a+4) \quad 18 \ (10c-7)(100c^2+70c+49)$$

$$19 \ 2(2a-3b)^3 \quad 20 \ \left(\frac{x}{2}-2\right)\left(\frac{x^2}{4}+x+4\right)$$

$$21 \ \left(\frac{a}{3}+\frac{b}{4}\right)\left(\frac{a^2}{9}-\frac{ab}{12}+\frac{b^2}{16}\right) \quad 22 \ (b-c)(b^2+bc+c^2+1)$$

$$23 \ (3y+4a)(9y^2-12ay+16a^2)$$

$$24 \ (b+2a)(b^2-2ab+4a^2-c)$$

$$25 \ (1-.2b)(.01+.02b+.04b^2) \quad 26 \ x(x^2-12x+48)$$

$$27 \ \left(\frac{2}{a}+\frac{6}{b}\right)\left(\frac{4}{a^2}-\frac{12}{ab}+\frac{36}{b^2}\right) \quad 28 \ \frac{4\pi}{3}(R-5)(R^2+5R+25)$$

$$29 \ (a-m^2n^3)(a^2+am^2n^3+m^6n^6)$$

$$30 \ (x-2)(x+2)(2x+1)(4x^2-2x+1)$$

15. Factorisation review

$$1 \ 4(2a+3) \quad 2 \ b^3(1-b) \quad 3 \ (c-5)(c+5) \quad 4 \ (x+6)(x+2)$$

$$5 \ (b+c)(a-1) \quad 6 \ 3(a+1)(a-7) \quad 7 \ (a-10)(a^2+10a+100)$$

$$8 \ 15(b-2)(b+2) \quad 9 \ 2(3x+1)(x+5) \quad 10 \ (a-b)(x+y)$$

$$11 \ (x-1)(x^2+a) \quad 12 \ 5(5x-4)(x-1) \quad 13 \ \left(\frac{y}{4}+\frac{x}{2}\right)\left(\frac{y}{4}-\frac{x}{2}\right)$$

$$14 \ 2a^2b(4ab-5b^3+2a^3) \quad 15 \ 8(a+4)(a^2-4a+16)$$

$$16 \ (a-1)^2 \quad 17 \ (2x-3y^2)(4x^2+16xy^2+9y^4) \quad 18 \ (x-y)^2$$

$$19 \ 2(3a-1)(2a+3) \quad 20 \ 3(c-24)(c+3)$$

$$21 \ 2(b-0.2)(b+0.2) \quad 22 \ (x+5)(x-5-a)$$

$$23 \ (7x+3)(5-3x) \quad 24 \ 5(5-2b)(5+2b)$$

$$25 \ (1-a)(1+a)(1+ab) \quad 26 \ (a-b)(y-5)$$

$$27 \ 7(3c+5b)(c-b) \quad 28 \ (a^2b-c^3)(a^2b+c^3)$$

$$29 \ 4(9a-5)(2a+3) \quad 30 \ 2\pi R(R+H) \quad 31 \ (c-d)^2(c+d)$$

$$32 \ 6(a+1)(a^2-a+1) \quad 33 \ (a+2y-1)(a-2y+7)$$

$$34 \ b^2(a-1)(a^2+a+1)(a^3+a^2+1) \quad 35 \ -3(2a-1)$$

$$36 \ 4x(2x-3)(x^2+5)$$

16. Simplifying algebraic expressions using factorisation

$$1 \ a+2 \quad 2 \ 4c-5 \quad 3 \ 5 \quad 4 \ \frac{1}{2} \quad 5 \ \frac{x-3}{4} \quad 6 \ \frac{a}{b} \quad 7 \ \frac{x-6}{x-1}$$

$$8 \ \frac{x}{y} \quad 9 \ \frac{1-3a}{a} \quad 10 \ \frac{-3}{2} \quad 11 \ \frac{b-c}{2} \quad 12 \ x+4 \quad 13 \ \frac{3a}{a+2}$$

$$14 \ -1 \quad 15 \ \frac{1}{fg-1} \quad 16 \ \frac{a-b}{a+b} \quad 17 \ \frac{x-2}{x+5}$$

$$18 \ \frac{1}{3} \quad 19 \ \frac{x^2+2x+4}{3} \quad 20 \ \frac{5y}{2x} \quad 21 \ \frac{a-1}{a+1}$$

$$22 \ \frac{x-4}{x+1} \quad 23 \ \frac{2x-3}{4x^2+6x+9} \quad 24 \ \frac{-(1+a)(1+a^2)}{2}$$

$$25 \ \frac{a-1}{a+b} \quad 26 \ x-2 \quad 27 \ \frac{a+2b}{b} \quad 28 \ \frac{x^2}{1+x+x^2}$$

$$29 \ \frac{a+c+m}{4} \quad 30 \ \frac{3(a+2)}{a+1}$$

17. Multiplication and division of algebraic fractions

$$1 \ \frac{2}{3} \quad 2 \ \frac{1}{4} \quad 3 \ \frac{3}{a+4} \quad 4 \ \frac{b+2}{2} \quad 5 \ \frac{a(x-3)}{10} \quad 6 \ \frac{a^2}{b}$$

$$7 \ \frac{1}{2(x^2-4x+16)} \quad 8 \ \frac{-a}{2b} \quad 9 \ \frac{4(x+1)}{(a+3)(x-5)}$$

$$10 \ \frac{-n^2(1+a)}{2} \quad 11 \ \frac{c+12}{c+2} \quad 12 \ \frac{2(c-e)}{c+e}$$

$$13 \ \frac{(x-y)(a-b)}{(a+b)(x+y)} \quad 14 \ -1 \quad 15 \ 1 \quad 16 \ \frac{2(a^2+ab+b^2)}{(c+3d)(a+b)}$$

$$17 \ -c(x-y) \quad 18 \ 2a+h \quad 19 \ \frac{b^2+3b+9}{2(a+2)(b+3)} \quad 20 \ \frac{2}{b}$$

18. Addition and subtraction of algebraic fractions

$$1 \ \frac{11a}{6} \quad 2 \ \frac{7a}{10} \quad 3 \ \frac{3a}{10} \quad 4 \ \frac{a+c}{b} \quad 5 \ \frac{-7}{6a} \quad 6 \ \frac{4b+5}{ab}$$

$$7 \ \frac{11x+17}{15} \quad 8 \ \frac{a}{12} \quad 9 \ \frac{5b+3a}{a^2b^2} \quad 10 \ \frac{53a}{60} \quad 11 \ \frac{-5b}{3}$$

$$12 \ \frac{5a-11}{12a} \quad 13 \ \frac{c+4a-2b}{abc} \quad 14 \ \frac{11x-5}{(x-3)(2x+1)}$$

$$15 \ \frac{2y}{(x-y)(x+y)} \quad 16 \ \frac{7}{2(x-3)} \quad 17 \ \frac{5a+12}{(a-2)(a+2)}$$

$$18 \ \frac{5x-2y}{x^3y^3} \quad 19 \ \frac{2c-2}{(c^2-9)(c+1)} \quad 20 \ \frac{1+2x+2y}{x(x+y)}$$

$$21 \ \frac{3-4a+4b}{(a-b)^2} \quad 22 \ \frac{5-2a-2b}{a^2-b^2} \quad 23 \ \frac{3x+13}{5(x^2-1)}$$

$$24 \ \frac{-2x-10}{x^2-9} \quad 25 \ \frac{7a+12}{a^2-25} \quad 26 \ \frac{a^2-2a+5}{a-2} \quad 27 \ \frac{4b}{1-b^2}$$

$$28 \ \frac{9x-2x^2}{x^3-16} \quad 29 \ \frac{5+3a-3b}{(a-b)^2} \quad 30 \ \frac{2x^2+3x+3}{(x^2-1)(x^2+x+1)}$$

19. Algebraic fractions review

$$1 \ 2(a+2) \quad 2 \ \frac{5}{2} \quad 3 \ a+b \quad 4 \ -x-3 \quad 5 \ \frac{x+1}{2} \quad 6 \ \frac{x+3}{x-3}$$

$$7 \ \frac{x-4}{2(x+2)} \quad 8 \ \frac{5b^2}{3a^2} \quad 9 \ 4 \quad 10 \ 1 \quad 11 \ \frac{3-x}{a} \quad 12 \ \frac{a}{12}$$

$$13 \ \frac{-2}{a+b} \quad 14 \ \frac{-c-d}{a-b} \quad 15 \ \frac{x-1}{x^2} \quad 16 \ \frac{2a}{3} \quad 17 \ \frac{a^2-b^2}{ab}$$

$$18 \ \frac{3a-22}{20} \quad 19 \ \frac{5}{3a} \quad 20 \ \frac{5d+3c}{c^2d^3} \quad 21 \ \frac{3x+9}{(x-2)(1+x)}$$

$$\begin{array}{llll}
 22 \frac{5}{2(x+6)} & 23 \frac{3x-1}{(x-1)^2} & 24 \frac{5x-23}{(x+2)(x-4)} & 25 \frac{19-4x}{x^2-16} \\
 26 \frac{12b}{9b^3-1} & 27 \frac{7x-15}{x^2-25} & 28 \frac{7x-36}{4x(x-4)} & 29 \frac{b}{a(a-b)} \\
 30 \frac{-4x^2-9x-36}{(x^2-9)(x^2+3x+9)}
 \end{array}$$

20. One step and two step equations

$$\begin{array}{llllll}
 1 a=8 & 2 a=22 & 3 a=8 & 4 a=48 & 5 a=92 & \\
 6 a=176 & 7 a=3\frac{3}{4} & 8 a=100 & 9 a=4 & 10 a=3\frac{2}{3} & \\
 11 a=1\frac{1}{5} & 12 a=\frac{8}{9} & 13 a=\frac{2}{3} & 14 a=-1\frac{4}{5} & & \\
 15 a=-4\frac{1}{3} & 16 a=-3 & 17 a=1 & 18 a=7\frac{1}{2} & & \\
 19 a=7 & 20 a=1\frac{2}{3} & 21 a= & 22 a=\frac{-2}{5} & & \\
 23 a=3 & 24 a=1\frac{1}{2} & 25 a= & 26 a=2 & & \\
 27 a=2\frac{1}{4} & 28 a=0.16 & 29 a=3\frac{1}{3} & 30 a=6 & &
 \end{array}$$

21. Three step equations

$$\begin{array}{llllll}
 1 a=1\frac{3}{4} & 2 a=4\frac{2}{5} & 3 a=\frac{7}{8} & 4 a=-7 & 5 a=-7 & \\
 6 a=-3 & 7 a=9\frac{1}{2} & 8 a=1\frac{1}{12} & 9 a=\frac{1}{6} & 10 a=1 & \\
 11 a=1 & 12 a=3 & 13 a=3 & 14 a=\frac{1}{3} & 15 a=\frac{1}{3} & \\
 16 a=1\frac{1}{2} & 17 a=1 & 18 a=2 & 19 a=1\frac{1}{5} & 20 a=-2\frac{5}{7} & \\
 21 a=\frac{-2}{3} & 22 a=3\frac{1}{4} & 23 a=1 & 24 a=-2 & 25 a=1 & \\
 26 a=5 & 27 a=5\frac{2}{3} & 28 a=3 & 29 a=\frac{2}{7} & 30 a=\frac{-11}{12} &
 \end{array}$$

22. Equations involving fractions

$$\begin{array}{llllll}
 1 a=11 & 2 a=1\frac{4}{5} & 3 a=14\frac{2}{3} & 4 a=\frac{2}{3} & 5 a=-10\frac{1}{4} & \\
 6 a=-3\frac{3}{4} & 7 a=1\frac{3}{14} & 8 a=\frac{-9}{11} & 9 a=7\frac{4}{7} & 10 a=3\frac{1}{3} & \\
 11 a=-9\frac{3}{4} & 12 a=-2\frac{1}{2} & 13 a=\frac{7}{8} & 14 a=\frac{1}{2} & & \\
 15 a=\frac{-7}{19} & 16 a=21 & 17 a=\frac{-3}{7} & 18 a=-4\frac{4}{5} & & \\
 19 a=-1 & 20 a=5\frac{1}{2} & 21 a=\frac{5}{19} & 22 a=1\frac{17}{27} & & \\
 23 a=\frac{-2}{3} & 24 a=\frac{2}{3} & 25 a=6\frac{1}{6} & 26 a=10\frac{1}{2} & & \\
 27 a=1 & 28 a=\frac{4}{5} & 29 a=-8 & 30 a=20 & &
 \end{array}$$

23. Exponential equations

$$\begin{array}{llllll}
 1 a=4 & 2 a=3 & 3 a=4 & 4 a=1 & 5 a=1 & \\
 6 a=4 & 7 a=0 & 8 a=1 & 9 a=-1\frac{1}{4} & 10 a=-\frac{1}{2} & \\
 11 a=-2 & 12 a=-5 & 13 a=7 & 14 a=8 & & \\
 15 a=4 & 16 a=-2 & 17 a=\frac{1}{2} & 18 a=10 & & \\
 19 a=4\frac{1}{2} & 20 a=-1 & 21 a=\frac{1}{2} & 22 a=-\frac{1}{2} & & \\
 23 a=-\frac{1}{6} & 24 a=-1\frac{1}{2} & 25 a=-1\frac{1}{2} & 26 a=3 & & \\
 27 a=1\frac{5}{6} & 28 a=2 & 29 a=\frac{1}{3} & 30 a=-2\frac{1}{2} & 31 a=5 & \\
 32 a=6 & 33 a=1 & 34 a=1\frac{3}{4} & 35 a=0 & 36 a=3 &
 \end{array}$$

24. Quadratic equations

$$\begin{array}{llllll}
 1 a=4, -1 & 2 a=-2, -3 & 3 a=-3, 2 & 4 a=18, -4 & & \\
 5 a=0, -7 & 6 a=0, 3 & 7 a=0, \frac{2}{3} & 8 a=0, \frac{1}{5} & & \\
 9 a=1, -2 & 10 a=4, -3 & 11 a=0, -2 & 12 a=4, -1 & & \\
 13 a=7, -\frac{1}{2} & 14 a=-3, -1\frac{2}{3} & 15 a=-2, 2\frac{6}{7} & & & \\
 16 a=4, -2\frac{2}{3} & 17 a=-4, -1\frac{1}{2} & 18 a=5, -5 & & & \\
 19 a=4, -4 & 20 a=1\frac{2}{3}, -1\frac{2}{3} & 21 a=\frac{3}{4}, -\frac{3}{4} & & & \\
 22 a=1\frac{1}{2}, -2\frac{1}{2} & 23 a=-7, 4 & 24 a=5, -2 & & & \\
 25 a=1 & 26 a=5\frac{1}{2}, -2 & 27 a=\frac{1}{2}, -\frac{1}{3} & 28 a=2\frac{1}{2} & & \\
 29 a=-2, 1\frac{1}{3} & 30 a=1, -2 & & & &
 \end{array}$$

25. Quadratic equations using the quadratic formula or completing the square

$$\begin{array}{llll}
 1 a=\frac{5 \pm \sqrt{61}}{2} & 2 a=-3 \pm \sqrt{7} & 3 a=\frac{-3 \pm \sqrt{37}}{2} & \\
 4 a=1 \pm \sqrt{7} & 5 a=3, -10 & 6 a=0, 9 & 7 a=\frac{3 \pm 3\sqrt{5}}{2} \\
 8 a=\frac{5 \pm \sqrt{97}}{4} & 9 a=\frac{-7 \pm \sqrt{37}}{6} & 10 a=\frac{7 \pm \sqrt{89}}{10} &
 \end{array}$$

$$\begin{array}{llll}
 11 a=\frac{3 \pm \sqrt{17}}{4} & 12 a=\frac{2 \pm \sqrt{2}}{2} & 13 a=1\frac{1}{2}, \frac{-2}{3} & \\
 14 a=\pm \sqrt{5} & 15 a=-1 \pm \sqrt{5} & 16 a=\frac{15 \pm \sqrt{265}}{10} & \\
 17 \text{ no solutions} & 18 a=-3 \pm \sqrt{13} & 19 a=5 \pm \sqrt{7} & \\
 20 a=2, 4 & 21 a=\pm 3 & 22 a=\pm 5 & 23 a=9, -1 \\
 24 a=0, 1, -1 & 25 a=0, -\frac{1}{2} & 26 a=\pm \sqrt{3} & \\
 27 a=1, -3\frac{2}{3} & 28 a=\frac{2}{3}, -5 & 29 a=1\frac{2}{3} & 30 a=2, 3
 \end{array}$$

26. Absolute value equations

$$\begin{array}{llllll}
 1 x=4, -4 & 2 x=2, -2 & 3 x=9, -1 & 4 x=-4, -8 & & \\
 5 x=-8 & 6 x=-2, 12 & 7 x=5, -8 & 8 x=3, -1\frac{2}{3} & & \\
 9 x=-1\frac{1}{5}, -2 & 10 x=8, -7 & 11 x=3\frac{1}{3}, -6 & & & \\
 12 x=2, -2 & 13 x=6, -2 & 14 x=\frac{-2}{3}, \frac{8}{11} & 15 x=-2 & & \\
 16 x=2\frac{9}{14}, \frac{-5}{26} & 17 x=2\frac{9}{11}, -1\frac{6}{11} & 18 x=3\frac{1}{14}, \frac{13}{14} & & & \\
 19 x=7, \frac{1}{3} & 20 x=-1\frac{6}{7}, -5\frac{2}{3} & 21 x=\frac{-1}{7}, -1 & & & \\
 22 x=3, -1 & 23 x=\frac{2}{3} & 24 x=-1 & 25 x=\frac{1}{6} & & \\
 26 x \geq 1\frac{2}{3} & 27 x=3, -3 & 28 x=-\frac{1}{2}, -\frac{3}{4} & & & \\
 29 \text{ no solutions} & 30 x=2, -6 & & & &
 \end{array}$$

27. Inequalities

$$\begin{array}{llllll}
 1 x < 5 & 2 x \geq -2 & 3 x \leq -2 & 4 x < 2 & 5 x > 1 & \\
 6 x > -4 & 7 x > 1 & 8 x \leq 1 & 9 x < -2\frac{1}{5} & 10 x < -7 & \\
 11 x < -2 & 12 x < -13 & 13 x \leq \frac{1}{3} & 14 x > 3\frac{1}{3} & & \\
 15 x \leq -7 & 16 x > 16\frac{1}{2} & 17 x > 3\frac{1}{7} & 18 x > -8 & & \\
 19 x < -19 & 20 x \geq -1 & 21 x < 2 & 22 x \geq -7 & & \\
 23 x < 1\frac{7}{9} & 24 x \leq 4 & 25 x \geq 31 & 26 x < 2\frac{25}{26} & & \\
 27 -2 < x < 2 & 28 x > -2\frac{5}{8} & 29 x > 1\frac{6}{13} & & & \\
 30 -3 < x < -1 & 31 x \geq -1\frac{15}{17} & & & &
 \end{array}$$

28. Absolute value inequalities

$$\begin{array}{llllll}
 1 -5 < x < 3 & 2 -1 \leq x \leq 4 & 3 x < 0, x > 12 & & & \\
 4 x < -2, x > \frac{2}{3} & 5 -\frac{1}{9} \leq x \leq 1 & 6 x < -4, x > 4 & & & \\
 7 -6 \leq x \leq 6 & 8 x < -14, x > 14 & 9 -1 < x < 1\frac{2}{3} & & & \\
 10 \text{ all real values of } x & 11 x < -5, x > 3 & 12 -4 \leq x \leq 8 & & & \\
 13 -3\frac{1}{2} < x < 6\frac{1}{2} & 14 x < -1\frac{1}{3}, x > 2 & 15 x < -2\frac{2}{3}, x > 0 & & & \\
 16 0 \leq x \leq 4 & 17 -\frac{5}{8} < x < 1\frac{5}{13} & 18 -17 < x < 13 & & & \\
 19 \text{ all real values of } x & 20 x < \frac{1}{2}, x > 3\frac{1}{2} & 21 -1\frac{1}{2} \leq x \leq \frac{1}{2} & & & \\
 22 x \leq 1\frac{3}{4}, x \geq 4\frac{1}{4} & 23 x < -1\frac{2}{3}, x > 3\frac{2}{3} & 24 -4 < x < \frac{-2}{3} & & & \\
 25 -\frac{1}{5} < x < 7 & & & & &
 \end{array}$$

29. Inequalities with the unknown in the denominator

$$\begin{array}{llllll}
 1 x < 2, x > 2\frac{1}{4} & 2 0 < x < \frac{1}{2} & 3 -3 < x < -1 & 4 x \leq \frac{2}{3}, x \geq 2 & & \\
 5 x < 0, x > 4 & 6 4 < x < 14 & 7 \frac{1}{2} \leq x < 2 & 8 x < 1, x > 3 & & \\
 9 x < -1, x > 2 & 10 3 < x < 8 & 11 -4\frac{1}{2} < x < 4 & & & \\
 12 0 < x < 1 & 13 x < -1, x > 3 & 14 \frac{-7}{10} < x < \frac{-1}{2} & & & \\
 15 -1 < x < 1, x > 3 & 16 x \leq -1, 0 < x < 2 & & & & \\
 17 x < -6, x > 1\frac{1}{2} & 18 -2 < x < -1\frac{2}{5} & & & &
 \end{array}$$

30. Quadratic inequalities

$$\begin{array}{llllll}
 1 -1 < x < 4 & 2 x < -2, x > 3 & 3 -4 < x < 2 & & & \\
 4 x < -5, x > -1 & 5 x \leq -2, x \geq 7 & 6 -6 < x < 0 & & & \\
 7 -8 < x < 9 & 8 4 \leq x \leq 9 & 9 -3 < x < 4 & 10 x < -5, x > 5 & & \\
 11 -6 < x < 6 & 12 -7 < x < 7 & 13 -3 < x < 5 & & & \\
 14 x < -10, x > -2 & 15 x < -4, x > 1\frac{2}{3} & 16 -5 < x < 5 & & & \\
 17 x < -1, x > 1 & 18 \text{ All } x \text{ except } x=1\frac{1}{2} & 19 -1\frac{1}{2} < x < \frac{2}{5} & & & \\
 20 -\frac{1}{2} < x < 7 & 21 x < 1\frac{2}{3}, x > 3 & 22 -1\frac{1}{2} < x < 1\frac{1}{2} & & & \\
 23 -3 < x < 5 & 24 x < \frac{2}{3}, x > 3\frac{1}{2} & 25 -6 < x < \frac{-1}{2} & & & \\
 26 -2 < x < 5 & 27 x < -1\frac{1}{3}, x > 1\frac{1}{2} & 28 \text{ no solutions} & & &
 \end{array}$$

29 no solutions 30 $0 < x < 7\frac{1}{2}$ 31 $x < -1\frac{1}{2}, x > -\frac{1}{6}$
 32 $x < -7, x > 7$ 33 $x < -3\sqrt{3}, x > 3\sqrt{3}$

31. Simultaneous equations - Elimination method

1 $x=1, y=1$ 2 $x=1, y=-1$ 3 $x=2, y=2$
 4 $x=4, y=-5$ 5 $a=2, b=3$ 6 $a=-1, b=-1$
 7 $a=5, b=2$ 8 $x=\frac{1}{2}, y=1$ 9 $x=-5, y=6$
 10 $c=2, d=5$ 11 $f=1, g=1$ 12 $x=2, y=-2$
 13 $x=\frac{1}{2}, y=3$ 14 $a=\frac{1}{3}, b=\frac{1}{4}$ 15 $a=\frac{1}{2}, b=-1$
 16 $c=1, d=-1$ 17 $x=1, y=1$ 18 $a=1, b=-3$
 19 $x=1, y=1$ 20 $x=\frac{1}{2}, y=-\frac{1}{3}$

32. Simultaneous equations - Substitution method

1 $x=1, y=2$ 2 $x=2, y=8$ 3 $x=-1, y=-1$
 4 $x=-1, y=3$ 5 $x=2, y=-1$ 6 $x=-2, y=-5$
 7 $x=2, y=1$ 8 $x=-1, y=5$ 9 $x=\frac{1}{4}, y=1\frac{1}{2}$
 10 $x=8, y=4$ 11 $x=3, y=-3$ and $x=4, y=0$
 12 $x=6, y=13$ and $x=-1, y=-1$
 13 $x=-2, y=8$ and $x=-6, y=20$
 14 $x=7, y=7$ and $x=-6, y=-6$
 15 $x=\frac{1}{2}, y=3\frac{1}{2}$ and $x=2, y=5$
 16 $x=4, y=1$ and $x=-1, y=-4$
 17 $x=\frac{1}{3}, y=6$ and $x=2, y=1$ 18 $x=3, y=-2$
 19 $x=1\frac{1}{3}, y=3\frac{2}{3}$ and $x=-2, y=-3$ 20 no solutions

33. Equations review

1 $x=4$ 2 $x=1\frac{2}{3}$ 3 $x=0$ 4 $x=\frac{3}{4}$ 5 $x=3$ 6 $x=5$
 7 $x=-2$ 8 $x=\frac{1}{2}$ 9 $x=3$ 10 $x=-1\frac{1}{2}$ 11 $x=\frac{12}{25}$
 12 $x=1\frac{7}{8}$ 13 $x=1\frac{16}{17}$ 14 $x=14\frac{3}{4}$ 15 $x=4\frac{1}{3}$
 16 $x=-1\frac{1}{2}$ 17 $x=4$ 18 $x=7, -1$ 19 $x=1\frac{1}{2}, 3$
 20 $x=0, \frac{1}{2}$ 21 $x=\frac{2 \pm \sqrt{10}}{3}$ 22 $x=2, -\frac{2}{3}$ 23 $x=1\frac{1}{4}, \frac{1}{12}$
 24 no solutions 25 $x=\frac{1}{3}$ 26 $x > -1$ 27 $x > 5$
 28 $x \leq -7, x \geq 8$ 29 $x < -\frac{5}{8}, x > \frac{1}{8}$ 30 $x < 0, x > 5$
 31 $-\frac{1}{2} < x \leq 1$ 32 $-5 < x < 5$ 33 $-7 < x < 9$
 34 $x < -1, x > 6$ 35 $x=1, y=2$ 36 $x=-1, y=-3$
 37 $x=2, y=4$ 38 $x=-\frac{1}{2}, y=2$
 39 $x=7, y=17$ and $x=-4, y=-5$

34. Simplifying surds

1 $2\sqrt{2}$ 2 $3\sqrt{3}$ 3 $8\sqrt{2}$ 4 $6\sqrt{2}$ 5 $3\sqrt{11}$ 6 $12\sqrt{2}$
 7 $2\sqrt{5}$ 8 $3\sqrt{6}$ 9 $9\sqrt{5}$ 10 $15\sqrt{2}$ 11 $10\sqrt{2}$
 12 $8\sqrt{11}$ 13 $40\sqrt{5}$ 14 $42\sqrt{3}$ 15 $x\sqrt{y}$ 16 $x=20$
 17 $x=18$ 18 $x=75$ 19 $x=112$ 20 $x=99$ 21 $x=3$
 22 $x=2$ 23 $x=9$ 24 $x=\frac{1}{25}$ 25 $x=\sqrt{29}$
 26 $x=3\sqrt{2}$ 27 $x=2\sqrt{5}$ 28 $x=\sqrt{65}$
 29 $3\sqrt{2}, \sqrt{16}, \sqrt{15}, 2\sqrt{3}, 2$ 30 $4\sqrt{3}, 7, 5\sqrt{2}, \sqrt{51}, 2\sqrt{13}$
 31 $3\sqrt{3}$ 32 x^2y^3 33 $25\sqrt{2}$

35. Addition and subtraction of surds

1 $2\sqrt{5}$ 2 0 3 $\sqrt{5} + \sqrt{3} + \sqrt{5}$ 4 $3\sqrt{5} - 7\sqrt{2}$
 5 $3\sqrt{2}$ 6 $4\sqrt{3}$ 7 $5\sqrt{2} + 5\sqrt{3}$ 8 0 9 $7 - 4\sqrt{5}$
 10 $5\sqrt{6} - 13\sqrt{11}$ 11 $8\sqrt{3}$ 12 $-8 - 4\sqrt{3}$
 13 $8\sqrt{7} - 8\sqrt{2}$ 14 $4 + 4\sqrt{2} + 4\sqrt{3}$ 15 $\sqrt{5} + 5\sqrt{2}$
 16 $4\sqrt{21} - \sqrt{33}$ 17 $12\sqrt{5} - 7\sqrt{3}$ 18 $-5\sqrt{2} - 10\sqrt{3}$
 19 $11\sqrt{2} - \sqrt{11}$ 20 $17\sqrt{3} - 18\sqrt{2}$ 21 $11\sqrt{2} - 2$
 22 $4\sqrt{a}$ 23 $-9\sqrt{3}$ 24 $2\sqrt{7}$ 25 $2\sqrt{3} + 3\sqrt{2} - 5$
 26 $9\sqrt{7} + 62$ 27 $8\sqrt{B}$ 28 $17\sqrt{10} + 10$
 29 $2x\sqrt{x} - y\sqrt{x}$ 30 $-9\sqrt{6}$

36. Multiplication of surds

1 $\sqrt{6}$ 2 5 3 6 4 $4\sqrt{3}$ 5 $9\sqrt{3}$ 6 20 7 $24\sqrt{2}$
 8 $3\sqrt{3}$ 9 10 10 7 11 12 12 54 13 12 14 $3\sqrt{2}$
 15 $10\sqrt{2}$ 16 $24\sqrt{3}$ 17 $5\sqrt{3} + 10$ 18 $28 - 14\sqrt{7}$
 19 $6 + 4\sqrt{3}$ 20 $12 - 4\sqrt{2}$ 21 $6\sqrt{2} + 14$
 22 $13\sqrt{3} + 11\sqrt{5}$ 23 $11\sqrt{2} - 17$ 24 $5\sqrt{10} - 41$
 25 $\sqrt{2} + 2\sqrt{3} + 6$ 26 $18\sqrt{2} - 7\sqrt{3}$ 27 $34 - 12\sqrt{3}$
 28 $20 - 7\sqrt{10} - 10\sqrt{5}$ 29 $-10\sqrt{3} - 2\sqrt{2}$ 30 $-7 - 13\sqrt{2}$

37. Binomial products involving surds

1 $\sqrt{15} + 7\sqrt{5} + 2\sqrt{3} + 14$ 2 $7 + 5\sqrt{3}$ 3 $42 + 24\sqrt{3}$
 4 $4\sqrt{5} - 7$ 5 $5\sqrt{3} + 5\sqrt{2} - \sqrt{6} - 2$ 6 $32 + \sqrt{35}$
 7 $34 - 18\sqrt{3}$ 8 $4 - 22\sqrt{6}$ 9 $a + 3\sqrt{ab} + 2b$
 10 $12 + 2\sqrt{35}$ 11 $5 + 2\sqrt{6}$ 12 $13 - 4\sqrt{3}$
 13 $49 + 12\sqrt{5}$ 14 $44 - 16\sqrt{7}$ 15 $29 - 4\sqrt{30}$
 16 $30 + 12\sqrt{6}$ 17 1 18 3 19 $a - b$
 20 $a + b + 2\sqrt{ab}$ 21 $a + 4b - 4\sqrt{ab}$ 22 $41 + 22\sqrt{3}$
 23 $16\sqrt{10} - 16$ 24 $-4 - 9\sqrt{5}$ 25 $3\sqrt{2} - 3$ 26 $9 + 3\sqrt{2}$
 27 $90 - 3\sqrt{6} + 24\sqrt{3} - 4\sqrt{2}$ 28 $2b - 2\sqrt{ab}$
 29 $37 + 30\sqrt{3}$ 30 $6 + 24\sqrt{6}$

38. Division of surds

1 $\sqrt{5}$ 2 $\sqrt{3}$ 3 2 4 $1\frac{1}{3}$ 5 $\sqrt{2}$ 6 5 7 1 8 $\frac{1}{2}$
 9 $\frac{3}{5}$ 10 2 11 $\frac{\sqrt{2}}{4}$ 12 $\sqrt{3}$ 13 1 14 $1\frac{1}{2}$ 15 10
 16 $\frac{\sqrt{3}}{2}$ 17 $4\sqrt{2}$ 18 33 19 $3\frac{1}{2}$ 20 $\frac{1}{2}$ 21 6 22 4
 23 $\frac{20}{\sqrt{2}} = 10\sqrt{2}$ 24 2 25 3 26 $\frac{2}{3}$ 27 3 28 $4\sqrt{6}$
 29 $6\frac{1}{4}$ 30 $\frac{9}{40}$ 31 1 32 a 33 $\frac{1}{4}$

39. Rationalising denominators

1 $\frac{5\sqrt{2}}{2}$ 2 $\frac{8\sqrt{3}}{3}$ 3 $\frac{\sqrt{14}}{7}$ 4 $\sqrt{2}$ 5 $\sqrt{2}$ 6 $\frac{3\sqrt{10}}{4}$
 7 $\frac{2\sqrt{3}}{3}$ 8 $\frac{3\sqrt{2}}{4}$ 9 $\frac{3\sqrt{3}}{2}$ 10 $\frac{\sqrt{10}}{2}$ 11 $\frac{3\sqrt{2}}{4}$
 12 $\frac{\sqrt{5}}{2}$ 13 $\sqrt{5} - \sqrt{3}$ 14 $3 + \sqrt{6}$ 15 $-2(2\sqrt{2} - 3)$
 16 $\frac{\sqrt{3}(\sqrt{3} + 1)}{2}$ 17 $\frac{5\sqrt{2}(\sqrt{5} - \sqrt{3})}{2}$ 18 $\sqrt{3}(\sqrt{5} + \sqrt{2})$
 19 $\frac{9(2\sqrt{2} + \sqrt{6})}{2}$ 20 $3 - 2\sqrt{2}$ 21 $6 - \sqrt{35}$
 22 $\frac{\sqrt{6} - \sqrt{2}}{4}$ 23 $\frac{\sqrt{3} + 3}{2}$ 24 $\frac{18\sqrt{3} - 11\sqrt{2}}{10}$
 25 $\frac{7 - 2\sqrt{6}}{75}$ 26 $a = \frac{1}{9}, b = -\frac{1}{10}$ 27 $a = 2, b = -1$
 28 $\frac{2\sqrt{5}}{3}$ 29 $\frac{8\sqrt{7}}{3}$ 30 $\frac{93\sqrt{10} + 294}{6} = \frac{31\sqrt{10} + 98}{2}$

40. Surds review

1 $5\sqrt{2}$ 2 $9\sqrt{7}$ 3 $x = 6\frac{1}{4}$ 4 $-6\sqrt{2}$ 5 $20\sqrt{3}$
 6 $7 + 6\sqrt{5}$ 7 $6\sqrt{3} - 17\sqrt{6}$ 8 $12 - 22\sqrt{2}$
 9 $2\sqrt{2} - 4\sqrt{5}$ 10 $(6 - a)\sqrt{a}$ 11 $\sqrt{30}$ 12 $5\sqrt{5}$
 13 $32\sqrt{3}$ 14 12 15 45 16 $54\sqrt{2}$ 17 $4 - 2\sqrt{2}$
 18 $7\sqrt{2} - 27$ 19 $3\sqrt{2} - 2 - 5\sqrt{3}$ 20 $\sqrt{3} - 4$
 21 47 22 $30 + 12\sqrt{6}$ 23 $108 - 48\sqrt{2}$ 24 $-8 + 2\sqrt{3}$
 25 $24\sqrt{10}$ 26 $\sqrt{6}$ 27 $\frac{2\sqrt{21}}{3}$ 28 8 29 $4\sqrt{2}$
 30 $3\sqrt{2}$ 31 $\frac{2\sqrt{2}}{3}$ 32 $\frac{4(3 + \sqrt{2})}{7}$ 33 $\frac{7 + 2\sqrt{6}}{5}$
 34 $\frac{5\sqrt{2} - \sqrt{6}}{4}$

Instant Lessons

Quality Ready-to-Use Resources

Each book in the **Instant Lessons** series is a collection of fresh and versatile worksheet activities which may be photocopied for student use. They are unique in providing the **highest quality learning outcomes with instant usability.**

Below is a checklist of the major uses and benefits of these books.

Versatile Uses		Teaching Benefits	
Supplementary classroom lessons	✓	Creative teaching ideas	✓
Homework activities	✓	Developing greater student interest and understanding	✓
Reinforcement of learning	✓	A fresh approach to a topic or lesson	✓
Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

ASSIGNMENTS IN SENIOR MATHS Book 1

This book contains a collection of pre-prepared worksheets for senior mathematics students of all abilities. The two-book series includes the most common topics from the various state syllabuses. Book 1 covers Basic Algebra & Indices, Factorisation, Algebraic Fractions and Surds. The activities are graded from simple revision questions to more difficult exercises for extension work. Answers are provided to make marking of student work easy and time-efficient.

Topics include:

Basic Algebra	One, Two & Three Step Equations
Negative Indices	Equations involving Fractions
Fractional Indices	Exponential Equations
Distributive Law	Quadratic Equations
Binomial Expansions	Inequations
Factorisation	Absolute Value Equations & Inequations
Monic & Non-monic Trinomials	Quadratic Inequations
Sum & Difference of Two Cubes	Simultaneous Equations
Simplifying Algebraic Expressions	Surds
Algebraic Fractions	Rationalising Denominators

Associated titles in the Instant Lessons series

- Assignments in Senior Maths (Book 2)
- Instant Lessons in Senior Maths (Books 1-5)
- Topics in Senior Maths (Books 1 & 2)

Copying Notice

Permission is given to schools or teachers who buy this book to make photocopies of all pages. The copies must be for internal school use only, and may not be given or sold to other educational institutions or to teachers from other institutions.